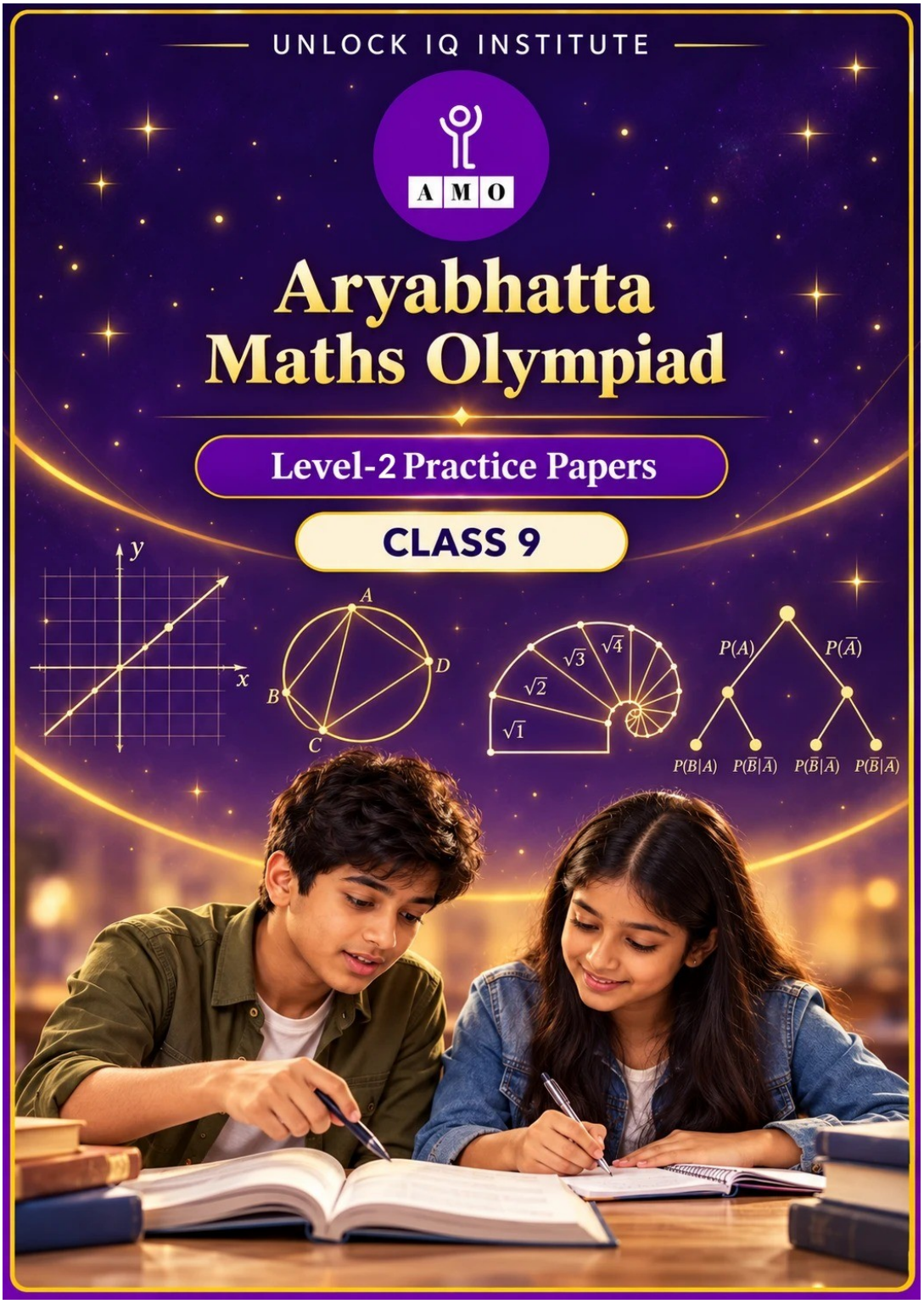



Set 2 · Practice Papers 6–10 · with answer keys & detailed solutions

| | |
|---|---|
| **Name:** _____ | **Class & Section:** _____ |
| **Roll No.:** _____ | **School:** _____ |
| **City:** _____ | **Date:** _____ |

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# ARYABHATTA MATHS OLYMPIAD

Level 2

## PRACTICE PAPER-6

# CLASS 9

| | |
|---|---|
| **Total Questions** | 30 multiple-choice questions |
| **Level** | Level-2 (final round): every question is Hard |
| **Time** | 45 minutes |
| **Arrangement** | Parts by chapter; the last Part has mixed problems joining two chapters |
| **Negative Marking** | No negative marking |
| **Calculator** | Not allowed |
| **Chapters** | Number System · Fractions, Decimals & Percentages · Algebra · Geometry · Mensuration · Data Handling, Statistics & Probability · Applied Mathematics · Mathematical Reasoning |

### Instructions

- Each question has four options, (A) to (D), with exactly one correct answer.
- The questions are grouped in Parts by chapter. The last Part has mixed problems, each joining two chapters.
- Every question is at Level-2 (Hard) standard: expect several steps or a key insight.
- There is no negative marking. Calculators are not allowed; use a separate sheet for rough work.
- Figures are not necessarily drawn to scale unless stated.

**For teachers**

- Figures are part of the questions and work when printed in black and white.
- Parts follow the AMO syllabus chapters. Chapters 2 and 7 are not part of the Class 9 syllabus, so the chapter numbers shown in the Part headings have gaps.
- The answer sheet and the answer key are at the end. Remove the answer key before the paper is attempted.

## Part 1 — Number System (Syllabus Chapter 1)

**Q1.** Which of these numbers is the largest?

**(A)** $\sqrt{2}$ **(B)** 1.4142 **(C)** 99/70 **(D)** 1.41

**Q2.** If $x = 3 + 2\sqrt{2}$, what is the value of $x^2 + 1/x^2$?

**(A)** 36 **(B)** 34 **(C)** 32 **(D)** $17 + 12\sqrt{2}$

## Part 2 — Fractions, Decimals & Percentages (Syllabus Chapter 3)

**Q3.** A herd of 2,000 deer grows by 25% each year through births, and then 200 deer are lost to predators at the end of each year. How many deer are there after 2 years?

**(A)** 2,725 **(B)** 2,600 **(C)** 2,675 **(D)** 2,850

**Q4.** A pond loses 5% of its water to evaporation every week. After 2 weeks it has 1,805 L. How much water did it have at the start?

**(A)** 2,000 L **(B)** 2,100 L **(C)** 1,995 L **(D)** 1,990 L

## Part 3 — Algebra (Syllabus Chapter 4)

**Q5.** If $x - 1/x = 3$, what is $x^4 + 1/x^4$?

**(A)** 119 **(B)** 121 **(C)** 81 **(D)** 123

**Q6.** Factorise $x^3 - 2x^2 - x + 2$ by grouping. What is the product of its three zeros?

**(A)** 2 **(B)** −1 **(C)** −2 **(D)** 0

**Q7.** A fraction becomes 1/2 when 1 is added to its numerator. It becomes 1/3 when 1 is subtracted from its denominator. What is the fraction?

**(A)** 1/4 **(B)** 2/5 **(C)** 1/3 **(D)** 3/8

**Q8.** How many three-digit numbers are divisible by 7?

**(A)** 129 **(B)** 128 **(C)** 127 **(D)** 142

## Part 4 — Geometry (Syllabus Chapter 5)

**Q9.** Which point lies on the straight line through (1, 5) and (2, 3)?

**(A)** (−2, 11) **(B)** (3, 2) **(C)** (0, 6) **(D)** (4, −2)

**Q10.** ABCD is a cyclic quadrilateral. Angle DBC = 55° and angle BAC = 45°. What is angle BCD?

**(A)** 100° **(B)** 80° **(C)** 55° **(D)** 45°

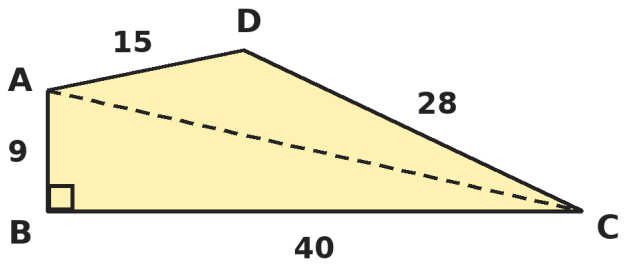
**Q11.** A chord of length 30 cm is drawn in a circle of radius 17 cm with centre O. What is the area of the triangle formed by O and the ends of the chord?

**(A)** 255 $cm^2$ **(B)** 240 $cm^2$ **(C)** 136 $cm^2$ **(D)** 120 $cm^2$

**Q12.** In triangle ABC, AB = AC = 13 cm and BC = 10 cm. G is the centroid. What is AG?

**(A)** 12 cm **(B)** 4 cm **(C)** 8 cm **(D)** 6 cm

## Part 5 — Mensuration (Syllabus Chapter 6)




**Q13.** In quadrilateral ABCD, AB = 9 m, BC = 40 m, CD = 28 m, DA = 15 m and angle ABC = 90°. What is its area?

**(A)** 180 $m^2$ **(B)** 126 $m^2$ **(C)** 306 $m^2$ **(D)** 352 $m^2$

**Q14.** A sector of radius 21 cm and angle 120° is rolled into a cone, without overlap. What is the radius of the base of the cone? (Take $\pi = 22/7$.)

**(A)** 21 cm **(B)** 7 cm **(C)** 14 cm **(D)** 3.5 cm

**Q15.** The radius of a sphere is increased by 50%. By what percentage does its volume increase?

**(A)** 150% **(B)** 237.5% **(C)** 125% **(D)** 337.5%

## Part 6 — Data Handling, Statistics & Probability (Syllabus Chapter 8)

**Q16.** Four fair coins are tossed together. What is the probability of getting more heads than tails?

**(A)** 1/2 **(B)** 11/16 **(C)** 3/8 **(D)** 5/16

**Q17.** The average age of 25 students is 40 months. When the teacher's age is included, the average rises by 1 month. What is the teacher's age in months?

**(A)** 66 **(B)** 41 **(C)** 65 **(D)** 26

**Q18.** Kunal scores 12, 0, 45, 102, 8, 45 and 30 runs in 7 innings. Which statement about the mean, median and mode is true?

**(A)** Mean > mode > median **(B)** Median > mean > mode

**(C)** Mean = median **(D)** Mode > mean > median

## Part 7 — Applied Mathematics (Syllabus Chapter 9)

**Q19.** Plant A is 12 cm tall and grows 1.5 cm each week. Plant B is 20 cm tall and grows 0.7 cm each week. After how many weeks will they be the same height?

**(A)** 10 **(B)** 8 **(C)** 12 **(D)** 16

**Q20.** 3 kg of rice and 2 kg of dal cost ₹420. 2 kg of rice and 3 kg of dal cost ₹480. What does 1 kg of dal cost?

**(A)** ₹60 **(B)** ₹120 **(C)** ₹180 **(D)** ₹96

**Q21.** An investment of ₹5,000 doubles every 6 years. Its value is checked only at the end of each 6-year period. After how many years is it first worth more than ₹1,00,000?

**(A)** 24 years **(B)** 30 years **(C)** 36 years **(D)** 20 years

## Part 8 — Mathematical Reasoning (Syllabus Chapter 10)

**Q22.** "If the diagonals of a quadrilateral are perpendicular, then it is a rhombus." Which shape is a counterexample to this claim?

**(A)** A square
**(B)** A rectangle that is not a square
**(C)** A rhombus
**(D)** A kite that is not a rhombus

**Q23.** Starting with 2, Pranav finds $2^{64}$ by squaring again and again: 2, $2^2$, $2^4$, ... How many multiplications does he make?

**(A)** 63 **(B)** 8 **(C)** 32 **(D)** 6

**Q24.** A Koch snowflake starts as an equilateral triangle of side 9 cm. At each step every side is replaced by 4 sides, each one-third as long. What is the perimeter after 2 steps?

**(A)** 36 cm **(B)** 64 cm **(C)** 48 cm **(D)** 27 cm

**Q25.** "In every triangle, the medians meet at a point that divides each median in the ratio 2 : 1." Which statement is a special case of this?

**(A)** In every quadrilateral, the diagonals meet at their midpoints.
**(B)** In every triangle, the altitudes meet at a point that divides each in the ratio 2 : 1.
**(C)** In an equilateral triangle, the centroid divides each altitude in the ratio 2 : 1.
**(D)** In a right triangle, the circumcentre is the midpoint of the hypotenuse.

## Part 9 — Mixed Problems (each joins two chapters)

**Q26.** Nandini adds all the odd numbers from 21 to 59: 21 + 23 + 25 + ... + 59. What total does she get?

**(A)** 780 **(B)** 820 **(C)** 900 **(D)** 800

**Q27.** A semicircle is drawn above the x-axis on the diameter from (0, 0) to (14, 0). What is the area inside the semicircle but outside the triangle with vertices (0, 0), (14, 0) and (7, 7)? (Take $\pi = 22/7$.)

**(A)** 28 **(B)** 77 **(C)** 49 **(D)** 126

**Q28.** The lines $y = 2x + 1$ and $y = -x + 7$ meet at P. The second line crosses the x-axis at Q. What is the distance PQ?

**(A)** $\sqrt{29}$ **(B)** $5\sqrt{2}$ **(C)** 7 **(D)** $4\sqrt{2}$

**Q29.** A two-digit number is chosen at random. What is the probability that its digits add up to 10?

**(A)** 1/10 **(B)** 1/9 **(C)** 9/100 **(D)** 1/11

**Q30.** Two TVs each cost ₹40,000. The first loses 25% of its value in the first year and 20% of its new value in the second. The second loses ₹9,000 each year. After 2 years, which is worth more, and by how much?

**(A)** The second, by ₹2,000 **(B)** They are worth the same

**(C)** The first, by ₹4,000 **(D)** The first, by ₹2,000

## Answer Sheet AMO Level 2 · Class 9 · Practice Paper 6

**Name:** **School:** **Roll No.:**

*Shade or tick one circle for each question.*

| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1. | A | B | C | D | 11. | A | B | C | D | 21. | A | B | C | D |
| 2. | A | B | C | D | 12. | A | B | C | D | 22. | A | B | C | D |
| 3. | A | B | C | D | 13. | A | B | C | D | 23. | A | B | C | D |
| 4. | A | B | C | D | 14. | A | B | C | D | 24. | A | B | C | D |
| 5. | A | B | C | D | 15. | A | B | C | D | 25. | A | B | C | D |
| 6. | A | B | C | D | 16. | A | B | C | D | 26. | A | B | C | D |
| 7. | A | B | C | D | 17. | A | B | C | D | 27. | A | B | C | D |
| 8. | A | B | C | D | 18. | A | B | C | D | 28. | A | B | C | D |
| 9. | A | B | C | D | 19. | A | B | C | D | 29. | A | B | C | D |
| 10. | A | B | C | D | 20. | A | B | C | D | 30. | A | B | C | D |

**For the parent or teacher**

**Correct: _____ / 30**

# Answer Key and Solutions AMO Level 2 · Class 9 · Practice Paper 6

## Quick answers

| | | | | | |
|---|---|---|---|---|---|
| 1 **C** | 6 **C** | 11 **D** | 16 **D** | 21 **B** | 26 **D** |
| 2 **B** | 7 **A** | 12 **C** | 17 **A** | 22 **D** | 27 **A** |
| 3 **C** | 8 **B** | 13 **C** | 18 **D** | 23 **D** | 28 **B** |
| 4 **A** | 9 **A** | 14 **B** | 19 **A** | 24 **C** | 29 **A** |
| 5 **A** | 10 **B** | 15 **B** | 20 **B** | 25 **C** | 30 **D** |

## Solutions

| Q | Part · level | Answer | Why? |
|---|---|---|---|
| **1** | Part 1 · Ch1 · Hard | **(C)** 99/70 | ***Hint:*** *Compare $(99/70)^2$ with 2.* $(99/70)^2 = 9{,}801/4{,}900$, which is more than 2 (9,800/4,900), so $99/70 > \sqrt{2} \approx 1.414214$. Also $\sqrt{2} > 1.4142 > 1.41$. So 99/70 is the largest; it is an old over-estimate of $\sqrt{2}$. |
| **2** | Part 1 · Ch1 · Hard | **(B)** 34 | ***Hint:*** *First find 1/x by multiplying $(3 + 2\sqrt{2})(3 - 2\sqrt{2})$.* $(3 + 2\sqrt{2})(3 - 2\sqrt{2}) = 9 - 8 = 1$, so $1/x = 3 - 2\sqrt{2}$ and $x + 1/x = 6$. Then $x^2 + 1/x^2 = (x + 1/x)^2 - 2 = 36 - 2 = 34$. (36 forgets to subtract 2; $17 + 12\sqrt{2}$ is $x^2$ alone.) |
| **3** | Part 2 · Ch3 · Hard | **(C)** 2,675 | ***Hint:*** *Apply the 25% growth first, then take away 200, and repeat.* Year 1: 2,000 × 1.25 = 2,500, then 2,500 − 200 = 2,300. Year 2: 2,300 × 1.25 = 2,875, then 2,875 − 200 = 2,675. (2,725 takes away both 200s at the end; 2,600 grows by a fixed 500 each year.) |
| **4** | Part 2 · Ch3 · Hard | **(A)** 2,000 L | ***Hint:*** *Each week the water is multiplied by 0.95. Undo it twice.* Start × 0.95 × 0.95 = 1,805, so Start = 1,805 ÷ 0.9025 = 2,000 L. (Adding back 10% of 1,805 gives a wrong 1,985.5.) |
| **5** | Part 3 · Ch4 · Hard | **(A)** 119 | ***Hint:*** *Square twice. Watch the middle terms.* $(x - 1/x)^2 = x^2 + 1/x^2 - 2 = 9$, so $x^2 + 1/x^2 = 11$. Squaring again: $x^4 + 1/x^4 + 2 = 121$, so $x^4 + 1/x^4 = 119$. |
| **6** | Part 3 · Ch4 · Hard | **(C)** −2 | ***Hint:*** *Group as $x^2(x - 2) - 1(x - 2)$.* $x^3 - 2x^2 - x + 2 = x^2(x - 2) - (x - 2) = (x - 2)(x^2 - 1) = (x - 2)(x - 1)(x + 1)$. The zeros are 2, 1 and −1; their product is −2. |
| **7** | Part 3 · Ch4 · Hard | **(A)** 1/4 | ***Hint:*** *Let the fraction be n/d and write two equations.* $(n + 1)/d = 1/2$ gives $d = 2n + 2$. $n/(d - 1) = 1/3$ gives $d = 3n + 1$. So $2n + 2 = 3n + 1$, $n = 1$ and $d = 4$. Check: $2/4 = 1/2$ and $1/3 = 1/3$. The fraction is 1/4. |
| **8** | Part 3 · Ch4 · Hard | **(B)** 128 | ***Hint:*** *They form an AP from the first to the last such number.* The first is 105 and the last is 994, with common difference 7. Number of terms = $(994 - 105)/7 + 1 = 127 + 1 = 128$. |
| **9** | Part 4 · Ch5 · Hard | **(A)** (−2, 11) | ***Hint:*** *Each step of 1 to the right takes the line 2 down.* Slope $= (3 - 5)/(2 - 1) = -2$, so $y = -2x + 7$. For $x = -2$: $y = 11$ ✓. $x = 3$ gives $y = 1$, $x = 0$ gives 7 and $x = 4$ gives −1, so the others are off the line. |

| Q | Part · level | Answer | Why? |
|---|---|---|---|
| **10** | Part 4 · Ch5 · Hard | **(B)** 80° | ***Hint:*** *Angles in the same segment are equal: angle DAC = angle DBC.* Angle DAC = angle DBC = 55° (same segment). So angle BAD = 45° + 55° = 100°. Opposite angles of a cyclic quadrilateral add to 180°, so angle BCD = 80°. |
| **11** | Part 4 · Ch5 · Hard | **(D)** 120 $cm^2$ | ***Hint:*** *The perpendicular from O bisects the chord.* The distance from O to the chord is $\sqrt{(17^2 - 15^2)}$ = 8 cm. Area = ½ × 30 × 8 = 120 $cm^2$. (255 uses the radius as the height.) |
| **12** | Part 4 · Ch5 · Hard | **(C)** 8 cm | ***Hint:*** *The median from A is also the altitude.* The median AD meets BC at its midpoint and is perpendicular to it, so AD = $\sqrt{(13^2 - 5^2)}$ = 12 cm. AG = ⅔ × 12 = 8 cm. |
| **13** | Part 5 · Ch6 · Hard | **(C)** 306 $m^2$ | ***Hint:*** *Find AC, then use Heron's formula for triangle ACD.* AC = $\sqrt{(9^2 + 40^2)}$ = 41 m. Triangle ABC = ½ × 9 × 40 = 180 $m^2$. Triangle ACD (41, 28, 15): s = 42, area = $\sqrt{(42 \times 1 \times 14 \times 27)}$ = 126 $m^2$. Total = 306 $m^2$. |
| **14** | Part 5 · Ch6 · Hard | **(B)** 7 cm | ***Hint:*** *The arc of the sector becomes the circumference of the base.* Arc length = (120/360) × 2 × 22/7 × 21 = 44 cm. So 2πr = 44 and r = 7 cm. The slant height of the cone is 21 cm. |
| **15** | Part 5 · Ch6 · Hard | **(B)** 237.5% | ***Hint:*** *Volume depends on $r^3$.* New volume = $1.5^3$ = 3.375 times the old one, an increase of 237.5%. (337.5% is the new volume as a percentage of the old one.) |
| **16** | Part 6 · Ch8 · Hard | **(D)** 5/16 | ***Hint:*** *More heads than tails means 3 or 4 heads. There are 16 equally likely outcomes.* 3 heads: 4 outcomes (the tail can be any one of the 4 coins). 4 heads: 1 outcome. Total 5 of 16, so 5/16. (1/2 ignores the 6 outcomes with 2 heads and 2 tails; 11/16 is "at least as many heads as tails".) |
| **17** | Part 6 · Ch8 · Hard | **(A)** 66 | ***Hint:*** *Compare the totals for 25 people and 26 people.* Students' total = 25 × 40 = 1,000. With the teacher: 26 × 41 = 1,066. Teacher = 66 months. (41 forgets that every student's share also rose by 1.) |
| **18** | Part 6 · Ch8 · Hard | **(D)** Mode > mean > median | ***Hint:*** *Order the scores first.* In order: 0, 8, 12, 30, 45, 45, 102. Median = 30, mode = 45, mean = 242/7 ≈ 34.6. So mode > mean > median. |
| **19** | Part 7 · Ch9 · Hard | **(A)** 10 | ***Hint:*** *Plant A gains 0.8 cm on plant B each week.* 12 + 1.5w = 20 + 0.7w gives 0.8w = 8, so w = 10 weeks. Both are then 27 cm tall. |
| **20** | Part 7 · Ch9 · Hard | **(B)** ₹120 | ***Hint:*** *Add and subtract the two equations.* Adding: 5r + 5d = 900, so r + d = 180. Subtracting: d − r = 60. So d = 120 and r = 60. Dal costs ₹120 per kg. |
| **21** | Part 7 · Ch9 · Hard | **(B)** 30 years | ***Hint:*** *Keep doubling 5,000 and count the doublings.* 5,000 → 10,000 → 20,000 → 40,000 → 80,000 → 1,60,000. After 4 doublings (24 years) it is ₹80,000; after 5 doublings (30 years) it is ₹1,60,000. So 30 years. (20 years comes from 1,00,000 ÷ 5,000 = 20.) |

| Q | Part · level | Answer | Why? |
|---|---|---|---|
| **22** | Part 8 · Ch10 · Hard | **(D)** A kite that is not a rhombus | ***Hint:*** *A counterexample must have perpendicular diagonals but not be a rhombus.* A kite with unequal pairs of sides has perpendicular diagonals but is not a rhombus, so it disproves the claim. A square and a rhombus agree with the claim, and a rectangle that is not a square does not have perpendicular diagonals. |
| **23** | Part 8 · Ch10 · Hard | **(D)** 6 | ***Hint:*** *Each squaring doubles the power.* Powers: 1 → 2 → 4 → 8 → 16 → 32 → 64. That is 6 squarings. (Multiplying by 2 each time would take 63 multiplications.) |
| **24** | Part 8 · Ch10 · Hard | **(C)** 48 cm | ***Hint:*** *Each step multiplies the perimeter by 4/3.* Start: 27 cm. After one step: 27 × 4/3 = 36 cm. After two steps: 36 × 4/3 = 48 cm. |
| **25** | Part 8 · Ch10 · Hard | **(C)** In an equilateral triangle, the centroid divides each altitude in the ratio 2 : 1. | ***Hint:*** *Look for a smaller family of triangles where the medians are something else as well.* In an equilateral triangle each median is also an altitude, so the theorem gives the statement about the equilateral triangle. The quadrilateral and altitude statements are false in general, and the right-triangle statement is a different theorem. |
| **26** | Part 9 · Mixed · Ch4 + Ch10 · Hard | **(D)** 800 | ***Hint:*** *$1 + 3 + \ldots + (2n - 1) = n^2$. Take away the odd numbers below 21.* $1 + 3 + \ldots + 59$ is the sum of the first 30 odd numbers: $30^2 = 900$. The odd numbers up to 19 are the first 10: $10^2 = 100$. So $21 + \ldots + 59 = 900 - 100 = 800$. (900 forgets to remove 1 to 19.) |
| **27** | Part 9 · Mixed · Ch5 + Ch6 · Hard | **(A)** 28 | ***Hint:*** *The point (7, 7) lies on the semicircle.* Semicircle: ½ × 22/7 × $7^2$ = 77. Triangle: ½ × 14 × 7 = 49. Difference = 28. |
| **28** | Part 9 · Mixed · Ch4 + Ch5 · Hard | **(B)** $5\sqrt{2}$ | ***Hint:*** *Solve the two equations together to find P.* $2x + 1 = -x + 7$ gives $x = 2$, $y = 5$, so P = (2, 5). Setting $y = 0$ in $y = -x + 7$ gives Q = (7, 0). PQ = $\sqrt{(5^2 + 5^2)} = 5\sqrt{2}$. |
| **29** | Part 9 · Mixed · Ch8 + Ch10 · Hard | **(A)** 1/10 | ***Hint:*** *There are 90 two-digit numbers.* The numbers are 19, 28, 37, 46, 55, 64, 73, 82, 91: 9 numbers out of 90. Probability = 9/90 = 1/10. |
| **30** | Part 9 · Mixed · Ch3 + Ch9 · Hard | **(D)** The first, by ₹2,000 | ***Hint:*** *Work out each value after two years.* First: 40,000 × 0.75 × 0.8 = ₹24,000. Second: 40,000 − 18,000 = ₹22,000. The first is worth ₹2,000 more. |

# ARYABHATTA MATHS OLYMPIAD

Level 2

## PRACTICE PAPER-7

# CLASS 9

| | |
|---|---|
| **Total Questions** | 30 multiple-choice questions |
| **Level** | Level-2 (final round): every question is Hard |
| **Time** | 45 minutes |
| **Arrangement** | Parts by chapter; the last Part has mixed problems joining two chapters |
| **Negative Marking** | No negative marking |
| **Calculator** | Not allowed |
| **Chapters** | Number System · Fractions, Decimals & Percentages · Algebra · Geometry · Mensuration · Data Handling, Statistics & Probability · Applied Mathematics · Mathematical Reasoning |

### Instructions

- Each question has four options, (A) to (D), with exactly one correct answer.
- The questions are grouped in Parts by chapter. The last Part has mixed problems, each joining two chapters.
- Every question is at Level-2 (Hard) standard: expect several steps or a key insight.
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- Figures are not necessarily drawn to scale unless stated.

**For teachers**

- Figures are part of the questions and work when printed in black and white.
- Parts follow the AMO syllabus chapters. Chapters 2 and 7 are not part of the Class 9 syllabus, so the chapter numbers shown in the Part headings have gaps.
- The answer sheet and the answer key are at the end. Remove the answer key before the paper is attempted.

## Part 1 — Number System (Syllabus Chapter 1)

**Q1.** Ishita inserts three rational numbers between 1/4 and 1/3 so that all five numbers are equally spaced. What is the middle number she inserts?

**(A)** 13/48 **(B)** 5/16 **(C)** 2/7 **(D)** 7/24

**Q2.** Exactly one of these numbers cannot be written as a fraction p/q. Which one?

**(A)** 0.123123123… **(B)** 22/7

**(C)** 0.1011011101111… (one more 1 in each block) **(D)** √1.44

## Part 2 — Fractions, Decimals & Percentages (Syllabus Chapter 3)

**Q3.** Scheme X pays ₹20,000 in year 1 and ₹1,000 more each year. Scheme Y pays ₹20,000 in year 1 and 5% more each year. Over the first three years, which pays more in total, and by how much?

**(A)** Scheme Y, by ₹50 **(B)** Scheme X, by ₹50

**(C)** They pay the same **(D)** Scheme Y, by ₹1,050

**Q4.** Rohit drives 120 km at 60 km/h and the next 120 km at 40 km/h. What is his average speed for the whole trip?

**(A)** 50 km/h **(B)** 45 km/h **(C)** 52 km/h **(D)** 48 km/h

## Part 3 — Algebra (Syllabus Chapter 4)

**Q5.** If $a^2 + b^2 = 25$, $a + b = 7$ and $a > b$, what is $a - b$?

**(A)** 7 **(B)** 1 **(C)** 5 **(D)** √24

**Q6.** $2x^2 + 7x + 3$ is factorised as $(ax + b)(cx + d)$, where a, b, c and d are positive whole numbers. What is the sum of the two factors?

**(A)** 3x + 4 **(B)** 3x + 7 **(C)** 2x + 4 **(D)** x + 4

**Q7.** How many points whose coordinates are both positive whole numbers lie on the line $3x + 4y = 48$?

**(A)** 4 **(B)** 5 **(C)** 2 **(D)** 3

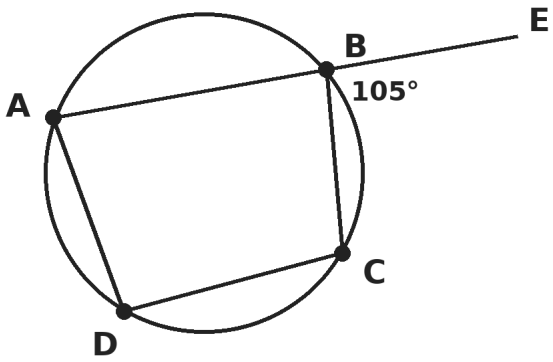
**Q8.** A sequence has $a_1 = 5$ and $a_{n+1} = a_n + 4$. Which is the first term greater than 200?

**(A)** 49th **(B)** 50th **(C)** 51st **(D)** 48th

## Part 4 — Geometry (Syllabus Chapter 5)

**Q9.** In triangle ABC, A = (0, 0), C = (5, 0) and B = (2, k) with k > 0. For which k is angle B a right angle?

**(A)** 6 **(B)** 3 **(C)** √6 **(D)** √3




**Q10.** ABCD is a cyclic quadrilateral. AB is extended to E, and angle CBE = 105°. What is angle ADC?

**(A)** 105° **(B)** 75° **(C)** 52.5° **(D)** 85°

**Q11.** In parallelogram ABCD, the diagonals AC = 16 cm and BD = 12 cm, and AB = 10 cm. What is its perimeter?

**(A)** 40 cm **(B)** 56 cm **(C)** 32 cm **(D)** 48 cm

**Q12.** The centroid of triangle ABC is G(3, 4), with A(1, 2) and B(6, 3). What is the length of the median from C?

**(A)** √10 **(B)** 3√10/2 **(C)** 2√10 **(D)** 3√10

## Part 5 — Mensuration (Syllabus Chapter 6)

**Q13.** A tricycle's front wheel has diameter 70 cm and each rear wheel has diameter 56 cm. Over a ride of 1.1 km, how many more complete turns does a rear wheel make than the front wheel? (Take $\pi = 22/7$.)

**(A)** 125 **(B)** 100 **(C)** 250 **(D)** 150

**Q14.** A cuboid has volume 1,440 $cm^3$ and a base 12 cm by 10 cm. What is its total surface area?

**(A)** 384 $cm^2$ **(B)** 528 $cm^2$ **(C)** 1,440 $cm^2$ **(D)** 768 $cm^2$

**Q15.** A square pyramid has base edge 12 cm and height 8 cm. What is its total surface area?

**(A)** 240 $cm^2$ **(B)** 528 $cm^2$ **(C)** 384 $cm^2$ **(D)** 192 $cm^2$

## Part 6 — Data Handling, Statistics & Probability (Syllabus Chapter 8)

**Q16.** Two fair dice are rolled. What is the probability that the product of the two numbers is a multiple of 6?

**(A)** 1/3 **(B)** 11/36 **(C)** 5/12 **(D)** 1/2

**Q17.** Six houses in a colony are priced at ₹40, 42, 44, 47 and 48 lakh and ₹2.5 crore. Which measure best describes a typical price, and what is its value?

**(A)** Mean, ₹78.5 lakh **(B)** Median, ₹45.5 lakh

**(C)** Mean, ₹45.5 lakh **(D)** Median, ₹44 lakh

**Q18.** Machine A makes 60% of a factory's bulbs and 2% of them are faulty. Machine B makes the rest and 5% of them are faulty. What is the probability that a bulb picked at random is faulty?

**(A)** 0.035 **(B)** 0.07 **(C)** 0.016 **(D)** 0.032

## Part 7 — Applied Mathematics (Syllabus Chapter 9)

**Q19.** A printer charges a fixed set-up fee plus a fixed amount per copy. 100 copies cost ₹650 and 250 copies cost ₹1,250. How much do 400 copies cost?

**(A)** ₹1,850 **(B)** ₹2,000 **(C)** ₹2,600 **(D)** ₹1,600

**Q20.** Simran and her mother have ages adding up to 50. In 5 years, her mother will be three times as old as Simran. How old was her mother when Simran was born?

**(A)** 40 **(B)** 30 **(C)** 35 **(D)** 25

**Q21.** On a map grid (1 unit = 1 km), Uma is at (3, −1) and Arnav at (−5, 7). They walk in straight lines to meet at the midpoint. How far does each walk?

**(A)** $8\sqrt{2}$ km **(B)** $4\sqrt{2}$ km **(C)** 8 km **(D)** 4 km

## Part 8 — Mathematical Reasoning (Syllabus Chapter 10)

**Q22.** Which of these statements is FALSE?

**(A)** The sum of a rational number and an irrational number is always irrational.

**(B)** The product of a non-zero rational number and an irrational number is always irrational.

**(C)** The sum of two irrational numbers is always irrational.

**(D)** The square of an irrational number can be rational.

**Q23.** An algorithm sets S = 0. For i = 1, 2, …, 10 it adds i to S when i is odd and subtracts i when i is even. What is S at the end?

**(A)** 5 **(B)** −5 **(C)** 0 **(D)** −55

**Q24.** In the Sierpinski triangle, each step replaces every shaded triangle by 3 smaller shaded ones. Starting from 1 triangle, after which step are there first more than 500 shaded triangles?

**(A)** 5 **(B)** 7 **(C)** 500 **(D)** 6

**Q25.** To prove there are infinitely many primes, assume there are only finitely many: $p_1, p_2, \ldots, p_k$. Let $N = p_1 \times p_2 \times \ldots \times p_k + 1$. Which fact gives the contradiction?

**(A)** N leaves remainder 1 when divided by every $p_i$

**(B)** N is always even

**(C)** N is always prime

**(D)** N is always a perfect square

## Part 9 — Mixed Problems (each joins two chapters)

**Q26.** A sequence starts with $a_1 = 1$ and follows the recursive rule $a_{n+1} = 2a_n + 3$. Which is the first term greater than 1,000?

**(A)** 8th **(B)** 9th **(C)** 10th **(D)** 7th

**Q27.** A field has corners (0, 0), (20, 0), (20, 14) and (0, 14), in metres. A circular pond of radius 7 m has its centre at (10, 7). What area of the field is not covered by the pond? (Take $\pi = 22/7$.)

**(A)** 280 $m^2$ **(B)** 154 $m^2$ **(C)** 126 $m^2$ **(D)** 434 $m^2$

**Q28.** The lines $y = 3x - 2$ and $y = x + 4$ form a triangle with the y-axis. What is its area?

**(A)** 18 **(B)** 6 **(C)** 9 **(D)** 12

**Q29.** In a class, the probability that a randomly chosen student is a boy is 3/5. When 6 more girls join, the probability becomes 1/2. How many students were in the class at first?

**(A)** 30 **(B)** 36 **(C)** 24 **(D)** 18

**Q30.** The price of petrol rises by 25% and then falls by 4%. By what percentage must Naveen reduce the amount of petrol he buys so that his spending stays the same?

**(A)** 21% **(B)** 20% **(C)** 17% **(D)** 16 2/3 %

## Answer Sheet AMO Level 2 · Class 9 · Practice Paper 7

**Name:** **School:** **Roll No.:**

*Shade or tick one circle for each question.*

| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1. | A | B | C | D | 11. | A | B | C | D | 21. | A | B | C | D |
| 2. | A | B | C | D | 12. | A | B | C | D | 22. | A | B | C | D |
| 3. | A | B | C | D | 13. | A | B | C | D | 23. | A | B | C | D |
| 4. | A | B | C | D | 14. | A | B | C | D | 24. | A | B | C | D |
| 5. | A | B | C | D | 15. | A | B | C | D | 25. | A | B | C | D |
| 6. | A | B | C | D | 16. | A | B | C | D | 26. | A | B | C | D |
| 7. | A | B | C | D | 17. | A | B | C | D | 27. | A | B | C | D |
| 8. | A | B | C | D | 18. | A | B | C | D | 28. | A | B | C | D |
| 9. | A | B | C | D | 19. | A | B | C | D | 29. | A | B | C | D |
| 10. | A | B | C | D | 20. | A | B | C | D | 30. | A | B | C | D |

**For the parent or teacher**

**Correct: _____ / 30**

# Answer Key and Solutions AMO Level 2 · Class 9 · Practice Paper 7

## Quick answers

| | | | | | |
|---|---|---|---|---|---|
| 1 **D** | 6 **A** | 11 **A** | 16 **C** | 21 **B** | 26 **B** |
| 2 **C** | 7 **D** | 12 **B** | 17 **B** | 22 **C** | 27 **C** |
| 3 **A** | 8 **B** | 13 **A** | 18 **D** | 23 **B** | 28 **C** |
| 4 **D** | 9 **C** | 14 **D** | 19 **A** | 24 **D** | 29 **A** |
| 5 **B** | 10 **A** | 15 **C** | 20 **B** | 25 **A** | 30 **D** |

## Solutions

| Q | Part · level | Answer | Why? |
|---|---|---|---|
| **1** | Part 1 · Ch1 · Hard | **(D)** 7/24 | ***Hint:*** *The gap 1/3 − 1/4 = 1/12 is split into 4 equal parts.* Each part is 1/48. The three numbers are 13/48, 14/48 and 15/48. The middle one is 14/48 = 7/24, which is also the average of 1/4 and 1/3. (2/7 lies between them but is not equally spaced.) |
| **2** | Part 1 · Ch1 · Hard | **(C)** 0.1011011101111… (one more 1 in each block) | ***Hint:*** *A rational number has a decimal that ends or repeats a fixed block.* 0.123123… repeats, 22/7 is a fraction and √1.44 = 1.2. The decimal 0.1011011101111… never ends and never repeats a fixed block, so it is irrational. |
| **3** | Part 2 · Ch3 · Hard | **(A)** Scheme Y, by ₹50 | ***Hint:*** *Write the three yearly payments for each scheme.* X: 20,000 + 21,000 + 22,000 = ₹63,000. Y: 20,000 + 21,000 + 22,050 = ₹63,050. Y pays ₹50 more (only in year 3 does growth on growth appear). |
| **4** | Part 2 · Ch3 · Hard | **(D)** 48 km/h | ***Hint:*** *Average speed = total distance ÷ total time.* Times: 2 h and 3 h, total 5 h for 240 km. Average speed = 240 ÷ 5 = 48 km/h. (50 is the plain average of the speeds, which gives the wrong weight to each part.) |
| **5** | Part 3 · Ch4 · Hard | **(B)** 1 | ***Hint:*** $(a - b)^2 = 2(a^2 + b^2) - (a + b)^2$. $(a - b)^2 = 2 \times 25 - 49 = 1$, so $a - b = 1$ (positive since $a > b$). Check: $a = 4$, $b = 3$. |
| **6** | Part 3 · Ch4 · Hard | **(A)** 3x + 4 | ***Hint:*** *Split 7x into two terms whose coefficients multiply to 2 × 3 = 6.* $2x^2 + 7x + 3 = 2x^2 + 6x + x + 3 = 2x(x + 3) + 1(x + 3) = (2x + 1)(x + 3)$. The sum is 3x + 4. |
| **7** | Part 3 · Ch4 · Hard | **(D)** 3 | ***Hint:*** *4y = 48 − 3x, so 3x must leave a multiple of 4. What does that say about x?* 48 − 3x must be a positive multiple of 4, so x must be a multiple of 4 and less than 16. x = 4, 8, 12 give y = 9, 6, 3. That is 3 points. (4 wrongly includes (16, 0), whose y-coordinate is not positive.) |
| **8** | Part 3 · Ch4 · Hard | **(B)** 50th | ***Hint:*** *Find an explicit rule for $a_n$.* $a_n = 5 + 4(n - 1) = 4n + 1$. $4n + 1 > 200$ gives $n > 49.75$, so $n = 50$ ($a_{50} = 201$, $a_{49} = 197$). |
| **9** | Part 4 · Ch5 · Hard | **(C)** √6 | ***Hint:*** *Use Pythagoras with squared distances: $AB^2 + BC^2 = AC^2$.* $AB^2 = 4 + k^2$, $BC^2 = 9 + k^2$, $AC^2 = 25$. $4 + k^2 + 9 + k^2 = 25$ gives $2k^2 = 12$, so $k = \sqrt{6}$. |

| Q | Part · level | Answer | Why? |
|---|---|---|---|
| **10** | Part 4 · Ch5 · Hard | **(A)** 105° | ***Hint:*** *First find angle ABC, then use opposite angles of a cyclic quadrilateral.* Angle ABC = 180° − 105° = 75°. Opposite angles of a cyclic quadrilateral add to 180°, so angle ADC = 105°. (The exterior angle equals the interior opposite angle.) |
| **11** | Part 4 · Ch5 · Hard | **(A)** 40 cm | ***Hint:*** *The diagonals bisect each other: AO = 8, BO = 6.* $AO^2 + BO^2 = 64 + 36 = 100 = AB^2$, so the diagonals meet at right angles. A parallelogram with perpendicular diagonals is a rhombus, so all sides are 10 cm and the perimeter is 40 cm. |
| **12** | Part 4 · Ch5 · Hard | **(B)** $3\sqrt{10}/2$ | ***Hint:*** *First find C. The centroid divides each median in the ratio 2 : 1 from the vertex.* A + B + C = 3G, so C = (9 − 7, 12 − 5) = (2, 7). CG = $\sqrt{((3-2)^2 + (4-7)^2)} = \sqrt{10}$. CG is 2/3 of the median, so the median = $(3/2)\sqrt{10} = 3\sqrt{10}/2$. ($\sqrt{10}$ is only CG.) |
| **13** | Part 5 · Ch6 · Hard | **(A)** 125 | ***Hint:*** *Find the circumference of each wheel in cm, and change 1.1 km to cm.* Front circumference = 22/7 × 70 = 220 cm; rear = 22/7 × 56 = 176 cm. 1.1 km = 1,10,000 cm. Front: 1,10,000 ÷ 220 = 500 turns; rear: 1,10,000 ÷ 176 = 625 turns. Difference = 125. |
| **14** | Part 5 · Ch6 · Hard | **(D)** 768 $cm^2$ | ***Hint:*** *Find the height first.* Height = 1,440 ÷ 120 = 12 cm. TSA = 2(12 × 10 + 10 × 12 + 12 × 12) = 2(120 + 120 + 144) = 768 $cm^2$. (384 forgets to double.) |
| **15** | Part 5 · Ch6 · Hard | **(C)** 384 $cm^2$ | ***Hint:*** *The slant height of a face is $\sqrt{(8^2 + 6^2)}$.* Slant height = $\sqrt{(64 + 36)}$ = 10 cm. Four faces: 4 × ½ × 12 × 10 = 240 $cm^2$. Base = 144 $cm^2$. Total = 384 $cm^2$. |
| **16** | Part 6 · Ch8 · Hard | **(C)** 5/12 | ***Hint:*** *Count pairs whose product has both a factor 2 and a factor 3.* Counting the 36 pairs: products divisible by 6 come from pairs with a 6 (11 pairs) plus (2,3), (3,2), (3,4), (4,3): 15 pairs. Probability = 15/36 = 5/12. |
| **17** | Part 6 · Ch8 · Hard | **(B)** Median, ₹45.5 lakh | ***Hint:*** *One price is far larger than the rest. ₹2.5 crore = ₹250 lakh.* The mean is (40 + 42 + 44 + 47 + 48 + 250)/6 = 78.5 lakh, pulled up by one house. The median is (44 + 47)/2 = 45.5 lakh, which is typical of most houses. So the median, ₹45.5 lakh. |
| **18** | Part 6 · Ch8 · Hard | **(D)** 0.032 | ***Hint:*** *Draw a tree: machine first, then faulty or not.* P = 0.6 × 0.02 + 0.4 × 0.05 = 0.012 + 0.020 = 0.032. (0.035 averages 2% and 5% without weights.) |
| **19** | Part 7 · Ch9 · Hard | **(A)** ₹1,850 | ***Hint:*** *The extra 150 copies cost ₹600.* Per copy: 600 ÷ 150 = ₹4. Set-up fee: 650 − 400 = ₹250. 400 copies: 250 + 1,600 = ₹1,850. (₹2,600 scales ₹650 by 4, counting the fee four times.) |
| **20** | Part 7 · Ch9 · Hard | **(B)** 30 | ***Hint:*** *Let Simran be d years old now; her mother is 50 − d.* 55 − d = 3(d + 5) gives 40 = 4d, so d = 10 and the mother is 40. The mother was 40 − 10 = 30 when Simran was born. |
| **21** | Part 7 · Ch9 · Hard | **(B)** $4\sqrt{2}$ km | ***Hint:*** *The midpoint is ((3 − 5)/2, (−1 + 7)/2).* Midpoint = (−1, 3). From (3, −1): $\sqrt{(4^2 + 4^2)} = 4\sqrt{2}$ km. ($8\sqrt{2}$ km is the whole distance between them.) |

| Q | Part · level | Answer | Why? |
|---|---|---|---|
| **22** | Part 8 · Ch10 · Hard | **(C)** The sum of two irrational numbers is always irrational. | ***Hint:*** *A single counterexample is enough to show a statement is false.* √2 and −√2 are irrational but their sum is 0, which is rational, so "the sum of two irrational numbers is always irrational" is false. The others are true: rational + irrational stays irrational, a non-zero rational times an irrational stays irrational, and $(\sqrt{2})^2 = 2$ is rational. |
| **23** | Part 8 · Ch10 · Hard | **(B)** −5 | ***Hint:*** *Group the steps in pairs: (1 − 2), (3 − 4), …* Pairs (1 − 2), (3 − 4), …, (9 − 10) each give −1, and there are 5 pairs. S = −5. |
| **24** | Part 8 · Ch10 · Hard | **(D)** 6 | ***Hint:*** *After step n there are $3^n$ triangles.* $3^5 = 243$ and $3^6 = 729$. So after step 6 there are first more than 500. |
| **25** | Part 8 · Ch10 · Hard | **(A)** N leaves remainder 1 when divided by every $p_i$ | ***Hint:*** *Which primes can divide N?* N leaves remainder 1 on division by each $p_i$, so no $p_i$ divides N. But N > 1 has a prime factor, which must be a prime not in the list — a contradiction. (N need not itself be prime: 2 × 3 × 5 × 7 × 11 × 13 + 1 = 30,031 = 59 × 509.) |
| **26** | Part 9 · Mixed · Ch4 + Ch10 · Hard | **(B)** 9th | ***Hint:*** *Write out the terms; each one is double the last plus 3.* The terms are 1, 5, 13, 29, 61, 125, 253, 509, 1,021. The 8th term is 509 and the 9th is 1,021, so the 9th term is the first above 1,000. (Check: $a_n$ + 3 doubles each time, 4, 8, 16, …, so $a_n = 2^{n+1} - 3$.) |
| **27** | Part 9 · Mixed · Ch5 + Ch6 · Hard | **(C)** 126 $m^2$ | ***Hint:*** *Check that the pond fits inside the field.* The pond touches the top and bottom edges but stays inside. Field = 20 × 14 = 280 $m^2$. Pond = 22/7 × 49 = 154 $m^2$. Remaining = 126 $m^2$. |
| **28** | Part 9 · Mixed · Ch4 + Ch5 · Hard | **(C)** 9 | ***Hint:*** *Find where the lines meet each other and the y-axis.* The lines meet where 3x − 2 = x + 4, at (3, 7). They meet the y-axis at (0, −2) and (0, 4): a base of 6. Height = 3. Area = ½ × 6 × 3 = 9. |
| **29** | Part 9 · Mixed · Ch8 + Ch10 · Hard | **(A)** 30 | ***Hint:*** *The number of boys does not change.* Boys = 3n/5. After 6 girls join: (3n/5)/(n + 6) = 1/2, so 6n/5 = n + 6 and n = 30 (18 boys, 12 girls). |
| **30** | Part 9 · Mixed · Ch3 + Ch9 · Hard | **(D)** 16 2/3 % | ***Hint:*** *The price factor is 1.25 × 0.96.* The new price is 1.25 × 0.96 = 1.2 times the old. To spend the same, he must buy 1/1.2 = 5/6 of the old amount, a cut of 1/6 = 16 2/3 %. |

# ARYABHATTA MATHS OLYMPIAD

Level 2

## PRACTICE PAPER-8

# CLASS 9

| | |
|---|---|
| **Total Questions** | 30 multiple-choice questions |
| **Level** | Level-2 (final round): every question is Hard |
| **Time** | 45 minutes |
| **Arrangement** | Parts by chapter; the last Part has mixed problems joining two chapters |
| **Negative Marking** | No negative marking |
| **Calculator** | Not allowed |
| **Chapters** | Number System · Fractions, Decimals & Percentages · Algebra · Geometry · Mensuration · Data Handling, Statistics & Probability · Applied Mathematics · Mathematical Reasoning |

### Instructions

- Each question has four options, (A) to (D), with exactly one correct answer.
- The questions are grouped in Parts by chapter. The last Part has mixed problems, each joining two chapters.
- Every question is at Level-2 (Hard) standard: expect several steps or a key insight.
- There is no negative marking. Calculators are not allowed; use a separate sheet for rough work.
- Figures are not necessarily drawn to scale unless stated.

**For teachers**

- Figures are part of the questions and work when printed in black and white.
- Parts follow the AMO syllabus chapters. Chapters 2 and 7 are not part of the Class 9 syllabus, so the chapter numbers shown in the Part headings have gaps.
- The answer sheet and the answer key are at the end. Remove the answer key before the paper is attempted.

## Part 1 — Number System (Syllabus Chapter 1)

**Q1.** How many of the numbers 0.34, 0.3434, 0.335 and 0.344 lie strictly between 1/3 and 0.343434… (34 repeating)?

**(A)** 2 **(B)** 4 **(C)** 3 **(D)** 1

**Q2.** To mark √50 on a number line, Nitya draws a right triangle with whole-number legs $a \le b$ and hypotenuse √50. How many different pairs (a, b) can she use?

**(A)** 1 **(B)** 2 **(C)** 3 **(D)** 0

## Part 2 — Fractions, Decimals & Percentages (Syllabus Chapter 3)

**Q3.** A machine bought for ₹50,000 is worth ₹32,000 after 2 years. It lost the same percentage of its value each year. What is that yearly percentage?

**(A)** 18% **(B)** 36% **(C)** 20% **(D)** 16%

**Q4.** 3 kg of an alloy with 60% copper is melted with 2 kg of an alloy with 85% copper. What percentage of the new alloy is copper?

**(A)** 72.5% **(B)** 75% **(C)** 70% **(D)** 68%

## Part 3 — Algebra (Syllabus Chapter 4)

**Q5.** Three numbers satisfy $a + b + c = 0$, with $a = 7$ and $b = -3$. Without cubing each number, find $a^3 + b^3 + c^3$.

**(A)** 252 **(B)** −252 **(C)** 0 **(D)** 84

**Q6.** For $p(x) = 2x^3 - 3x^2 + ax - 6$, we know $p(2) = 4$. What is $p(-1)$?

**(A)** −14 **(B)** −8 **(C)** −2 **(D)** 14

**Q7.** Kiara has 30 coins, all ₹5 or ₹2, worth ₹105 in all. How many ₹5 coins does she have?

**(A)** 12 **(B)** 18 **(C)** 15 **(D)** 10

**Q8.** How many terms of the arithmetic progression 3, 7, 11, … are less than 500?

**(A)** 124 **(B)** 125 **(C)** 126 **(D)** 100

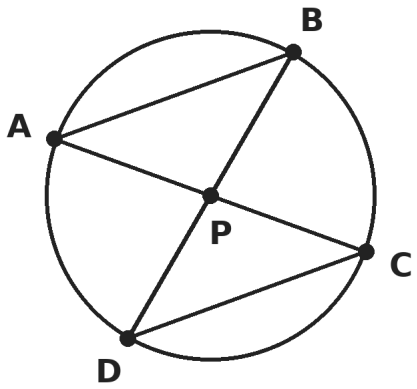
## Part 4 — Geometry (Syllabus Chapter 5)

**Q9.** The triangle has vertices A(−1, 0), B(3, 3) and C(3, −3). Its circumcentre lies on the x-axis. How far is the circumcentre from A?

**(A)** 4 **(B)** 5/2 **(C)** 25/8 **(D)** 3

**Q10.** A chord AB of length 8 cm is 3 cm from the centre of a circle. What is the greatest distance from the chord AB to a point on the circle?

**(A)** 8 cm **(B)** 5 cm **(C)** 3 cm **(D)** 2 cm




**Q11.** Chords AC and BD of a circle meet at P. Angle BAC = 40° and angle APD = 100°. What is angle ACD?

**(A)** 40° **(B)** 80° **(C)** 60° **(D)** 100°

**Q12.** A quadrilateral has perpendicular diagonals of lengths 8 cm and 10 cm. The midpoints of its sides are joined in order. What is the area of the new quadrilateral?

**(A)** 40 cm² **(B)** 80 cm² **(C)** 18 cm² **(D)** 20 cm²

## Part 5 — Mensuration (Syllabus Chapter 6)

**Q13.** A sector of a circle of radius 21 cm has a perimeter of 64 cm. What is its area? (Use π = 22/7.)

**(A)** 462 cm² **(B)** 346.5 cm² **(C)** 693 cm² **(D)** 231 cm²

**Q14.** The inside of a hemispherical dome of radius 14 m is to be painted at ₹50 per $m^2$. What is the cost? (Take $\pi = 22/7$.)

**(A)** ₹30,800 **(B)** ₹92,400 **(C)** ₹61,600 **(D)** ₹1,23,200

**Q15.** A cone has volume $96\pi$ $cm^3$ and base radius 6 cm. What is its curved surface area?

**(A)** $96\pi$ $cm^2$ **(B)** $60\pi$ $cm^2$ **(C)** $48\pi$ $cm^2$ **(D)** $36\pi$ $cm^2$

## Part 6 — Data Handling, Statistics & Probability (Syllabus Chapter 8)

**Q16.** A leap year is chosen at random. What is the probability that it has 53 Sundays?

**(A)** 1/7 **(B)** 3/7 **(C)** 53/366 **(D)** 2/7

**Q17.** The average of 25 numbers is 15.6. The average of the first 10 of them is 12. A new number is added to the other 15, and their average becomes 18.5. What is the new number?

**(A)** 18.5 **(B)** 34 **(C)** 26 **(D)** 22

| Marks | 10 | 20 | 30 | 40 | 50 |
|---|---|---|---|---|---|
| Students | 4 | 7 | 12 | 9 | 8 |

**Q18.** The table shows the marks of 40 students. What is the mean minus the median?

**(A)** 0 **(B)** 7.5 **(C)** −2.5 **(D)** 2.5

## Part 7 — Applied Mathematics (Syllabus Chapter 9)

**Q19.** A concert sells 300 tickets, some at ₹200 and the rest at ₹500, for ₹90,000 in all. How many ₹500 tickets were sold?

**(A)** 100 **(B)** 200 **(C)** 120 **(D)** 150

**Q20.** The amount of a medicine in the blood halves every 4 hours. A patient takes 400 mg at 8 AM and another 400 mg at 4 PM. How much is in the blood at midnight?

**(A)** 100 mg **(B)** 125 mg **(C)** 150 mg **(D)** 225 mg

**Q21.** A factory checks 200 items on Monday and finds 4 faulty, and 300 items on Tuesday and finds 9 faulty. What is the best estimate of the probability that an item is faulty?

**(A)** 2.5% **(B)** 3% **(C)** 2% **(D)** 2.6%

## Part 8 — Mathematical Reasoning (Syllabus Chapter 10)

**Q22.** How many whole numbers from 1 to 100 are counterexamples to the claim "If a number is divisible by 4 and by 6, then it is divisible by 24"?

**(A)** 4 **(B)** 8 **(C)** 3 **(D)** 5

**Q23.** Algorithm: start with 100; repeatedly replace the number by the whole-number part of half of it, until you reach 0. How many halvings are made?

**(A)** 6 **(B)** 7 **(C)** 8 **(D)** 50

**Q24.** In how many ways can a $2 \times 6$ board be covered completely by $2 \times 1$ dominoes?

**(A)** 8 **(B)** 13 **(C)** 21 **(D)** 12

**Q25.** Kunal claims: "Every odd number greater than 1 is either a prime or a perfect square." Which number is a counterexample to his claim?

**(A)** 9 **(B)** 25 **(C)** 13 **(D)** 15

## Part 9 — Mixed Problems (each joins two chapters)

**Q26.** Matchsticks form a grid of unit squares that is 2 squares tall and n squares long. For example, a $2 \times 1$ grid uses 7 sticks. How many unit squares are in the grid made with exactly 302 sticks?

**(A)** 60 **(B)** 100 **(C)** 150 **(D)** 120

**Q27.** Two parallel chords of lengths 16 cm and 12 cm lie on opposite sides of the centre of a circle of radius 10 cm. Their ends form a trapezium. What is its area?

**(A)** 196 $cm^2$ **(B)** 28 $cm^2$ **(C)** 56 $cm^2$ **(D)** 98 $cm^2$

**Q28.** The lines $y = 2x + 1$ and $y = 2x - 7$ cross the y-axis at P and Q and the x-axis at R and S. What is the area of the quadrilateral PRQS?

**(A)** 16 **(B)** 32 **(C)** 8 **(D)** 12

**Q29.** A number is chosen at random from 1 to 50. What is the probability that it is a multiple of 4 but not a multiple of 6?

**(A)** 6/25 **(B)** 4/25 **(C)** 1/5 **(D)** 2/25

**Q30.** A loan of ₹60,000 is repaid at ₹5,000 a month. Each month, before the payment, interest of 1% is also paid on the amount still owed. How much interest is paid in total?

**(A)** ₹3,900 **(B)** ₹7,200 **(C)** ₹3,600 **(D)** ₹4,200

## Answer Sheet AMO Level 2 · Class 9 · Practice Paper 8

**Name:** **School:** **Roll No.:**

*Shade or tick one circle for each question.*

| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1. | A | B | C | D | 11. | A | B | C | D | 21. | A | B | C | D |
| 2. | A | B | C | D | 12. | A | B | C | D | 22. | A | B | C | D |
| 3. | A | B | C | D | 13. | A | B | C | D | 23. | A | B | C | D |
| 4. | A | B | C | D | 14. | A | B | C | D | 24. | A | B | C | D |
| 5. | A | B | C | D | 15. | A | B | C | D | 25. | A | B | C | D |
| 6. | A | B | C | D | 16. | A | B | C | D | 26. | A | B | C | D |
| 7. | A | B | C | D | 17. | A | B | C | D | 27. | A | B | C | D |
| 8. | A | B | C | D | 18. | A | B | C | D | 28. | A | B | C | D |
| 9. | A | B | C | D | 19. | A | B | C | D | 29. | A | B | C | D |
| 10. | A | B | C | D | 20. | A | B | C | D | 30. | A | B | C | D |

**For the parent or teacher**

**Correct: _____ / 30**

# Answer Key and Solutions AMO Level 2 · Class 9 · Practice Paper 8

## Quick answers

| | | | | | |
|---|---|---|---|---|---|
| 1 **C** | 6 **A** | 11 **C** | 16 **D** | 21 **D** | 26 **D** |
| 2 **B** | 7 **C** | 12 **D** | 17 **C** | 22 **A** | 27 **A** |
| 3 **C** | 8 **B** | 13 **D** | 18 **D** | 23 **B** | 28 **A** |
| 4 **C** | 9 **C** | 14 **C** | 19 **A** | 24 **B** | 29 **B** |
| 5 **A** | 10 **A** | 15 **B** | 20 **B** | 25 **D** | 30 **A** |

## Solutions

| Q | Part · level | Answer | Why? |
|---|---|---|---|
| **1** | Part 1 · Ch1 · Hard | **(C)** 3 | ***Hint:*** *1/3 = 0.3333... Compare digit by digit.* 0.34, 0.3434 and 0.335 are all more than 0.3333... and less than 0.343434... (0.3434 < 0.343434...). 0.344 is more than 0.3434..., so it is outside. That is 3 numbers. |
| **2** | Part 1 · Ch1 · Hard | **(B)** 2 | ***Hint:*** *$a^2 + b^2 = 50$.* $a^2 + b^2 = 50$: (1, 7) gives 1 + 49 and (5, 5) gives 25 + 25. No other pair works. So 2 pairs. |
| **3** | Part 2 · Ch3 · Hard | **(C)** 20% | ***Hint:*** *The value is multiplied by the same factor twice: factor² = 32,000 ÷ 50,000.* factor² = 32,000/50,000 = 0.64, so the factor is 0.8: a 20% loss each year. Check: 50,000 × 0.8 = 40,000 and 40,000 × 0.8 = 32,000. (18% splits the total ₹18,000 loss as a percentage; 36% is the total fall over both years.) |
| **4** | Part 2 · Ch3 · Hard | **(C)** 70% | ***Hint:*** *Find the total mass of copper.* Copper = 0.6 × 3 + 0.85 × 2 = 1.8 + 1.7 = 3.5 kg out of 5 kg = 70%. (72.5% is the simple average of 60% and 85%.) |
| **5** | Part 3 · Ch4 · Hard | **(A)** 252 | ***Hint:*** *If $a + b + c = 0$, then $a^3 + b^3 + c^3 = 3abc$.* c = −4. Since a + b + c = 0, $a^3 + b^3 + c^3 = 3abc = 3 \times 7 \times (-3) \times (-4) =$ 252. Check: 343 − 27 − 64 = 252. |
| **6** | Part 3 · Ch4 · Hard | **(A)** −14 | ***Hint:*** *Use p(2) = 4 to find a first.* p(2) = 16 − 12 + 2a − 6 = 4 gives 2a = 6, a = 3. p(−1) = −2 − 3 − 3 − 6 = −14. |
| **7** | Part 3 · Ch4 · Hard | **(C)** 15 | ***Hint:*** *If all were ₹2 coins, they would be worth ₹60.* All ₹2 coins would make ₹60. Each ₹5 coin adds ₹3 more. (105 − 60) ÷ 3 = 15 coins of ₹5. |
| **8** | Part 3 · Ch4 · Hard | **(B)** 125 | ***Hint:*** *The nth term is 4n − 1.* 4n − 1 < 500 gives n < 125.25. So n = 1, 2, ..., 125: 125 terms (the 125th is 499). |
| **9** | Part 4 · Ch5 · Hard | **(C)** 25/8 | ***Hint:*** *Let the circumcentre be (x, 0) and make its distances to A and B equal.* $(x + 1)^2 = (x - 3)^2 + 9$ gives 2x + 1 = −6x + 18, so x = 17/8. Distance from A = 17/8 + 1 = 25/8. Check with B: $\sqrt{((3 - 17/8)^2 + 9)} = \sqrt{(49/64 + 576/64)} = 25/8$. (4 is the distance from A to the midpoint of BC.) |

| Q | Part · level | Answer | Why? |
|---|---|---|---|
| **10** | Part 4 · Ch5 · Hard | **(A)** 8 cm | ***Hint:*** *The farthest point lies on the line through the centre perpendicular to AB, on the far side.* Radius = $\sqrt{(4^2 + 3^2)} = 5$ cm. The farthest point is on the far side of the centre: 3 + 5 = 8 cm from the chord. (2 cm is the nearest point on the other arc.) |
| **11** | Part 4 · Ch5 · Hard | **(C)** 60° | ***Hint:*** *Angle APB = 180° − 100°. Then use angles in the same segment.* Angle APB = 80°, so in triangle APB, angle ABP = 180° − 40° − 80° = 60°. Angle ACD and angle ABD stand on the same arc AD, so angle ACD = 60°. |
| **12** | Part 4 · Ch5 · Hard | **(D)** 20 $cm^2$ | ***Hint:*** *The new shape has sides parallel to the diagonals and half as long.* By the midpoint theorem the new shape has sides 4 cm and 5 cm, parallel to the diagonals. The diagonals are perpendicular, so it is a rectangle: area 4 × 5 = 20 $cm^2$ (half the original area of 40 $cm^2$). |
| **13** | Part 5 · Ch6 · Hard | **(D)** 231 $cm^2$ | ***Hint:*** *Find the arc length first; the area of a sector is ½ × radius × arc.* Arc = 64 − 2 × 21 = 22 cm. Area = ½ × 21 × 22 = 231 $cm^2$. (462 forgets the ½.) |
| **14** | Part 5 · Ch6 · Hard | **(C)** ₹61,600 | ***Hint:*** *The curved surface of a hemisphere is $2\pi r^2$.* Curved surface = $2 \times 22/7 \times 14^2$ = 1,232 $m^2$. Cost = 1,232 × 50 = ₹61,600. (₹92,400 adds the floor; ₹1,23,200 uses a full sphere.) |
| **15** | Part 5 · Ch6 · Hard | **(B)** 60π $cm^2$ | ***Hint:*** *Find the height, then the slant height.* ⅓ × π × 36 × h = 96π gives h = 8 cm. Slant height = $\sqrt{(36 + 64)} = 10$ cm. Curved surface = π × 6 × 10 = 60π $cm^2$. |
| **16** | Part 6 · Ch8 · Hard | **(D)** 2/7 | ***Hint:*** *366 days = 52 weeks + 2 extra days.* The 2 extra days are a pair of consecutive days: (Sun, Mon), (Mon, Tue), …, (Sat, Sun) — 7 equally likely pairs. Two of them include Sunday, so the probability is 2/7. |
| **17** | Part 6 · Ch8 · Hard | **(C)** 26 | ***Hint:*** *Find the total of the other 15 numbers first.* Total of 25 = 25 × 15.6 = 390; first 10 = 120; other 15 = 270 (average 18). After adding x: 16 × 18.5 = 296, so x = 296 − 270 = 26. (18.5 is the new average, not the new number.) |
| **18** | Part 6 · Ch8 · Hard | **(D)** 2.5 | ***Hint:*** *The median is the average of the 20th and 21st marks in order.* Mean = (40 + 140 + 360 + 360 + 400)/40 = 1,300/40 = 32.5. The cumulative counts are 4, 11, 23, … so the 20th and 21st marks are both 30: median 30. Mean − median = 2.5. |
| **19** | Part 7 · Ch9 · Hard | **(A)** 100 | ***Hint:*** *If all were ₹200 tickets, the takings would be ₹60,000.* 300 tickets at ₹200 would give ₹60,000. Each ₹500 ticket adds ₹300 more. (90,000 − 60,000) ÷ 300 = 100 tickets at ₹500. |
| **20** | Part 7 · Ch9 · Hard | **(B)** 125 mg | ***Hint:*** *Find the amount just after the second dose, then halve it twice.* From 8 AM to 4 PM is 2 halvings: 400 → 200 → 100 mg. After the second dose: 500 mg. From 4 PM to midnight is 2 more halvings: 500 → 250 → 125 mg. (100 mg forgets the second dose.) |
| **21** | Part 7 · Ch9 · Hard | **(D)** 2.6% | ***Hint:*** *Combine all the data rather than averaging the two rates.* Altogether 13 faulty out of 500 items: 13/500 = 2.6%. (2.5% averages 2% and 3% without weighting by the number of items.) |

| Q | Part · level | Answer | Why? |
|---|---|---|---|
| **22** | Part 8 · Ch10 · Hard | **(A)** 4 | ***Hint:*** *Divisible by 4 and by 6 means divisible by 12.* Numbers divisible by 4 and by 6 are the multiples of 12: 12, 24, 36, 48, 60, 72, 84, 96. Those not divisible by 24 are 12, 36, 60 and 84. So there are 4 counterexamples. (8 counts every multiple of 12.) |
| **23** | Part 8 · Ch10 · Hard | **(B)** 7 | ***Hint:*** *100 → 50 → 25 → 12 → …* 100 → 50 → 25 → 12 → 6 → 3 → 1 → 0: 7 halvings. (This equals the number of binary digits of 100, which is 1100100.) |
| **24** | Part 8 · Ch10 · Hard | **(B)** 13 | ***Hint:*** *The left end is either one upright domino or two flat dominoes: W(n) = W(n − 1) + W(n − 2).* W(1) = 1, W(2) = 2, then 3, 5, 8, 13. So W(6) = 13. |
| **25** | Part 8 · Ch10 · Hard | **(D)** 15 | ***Hint:*** *A counterexample must be odd, not prime and not a perfect square.* $9 = 3^2$ and $25 = 5^2$ are perfect squares, and 13 is prime, so they fit the claim. $15 = 3 \times 5$ is odd, not prime and not a square, so it disproves the claim. |
| **26** | Part 9 · Mixed · Ch4 + Ch10 · Hard | **(D)** 120 | ***Hint:*** *Count horizontal sticks (3 rows of n) and vertical sticks (n + 1 columns of 2).* Sticks = 3n + 2(n + 1) = 5n + 2. Check n = 1: 7 ✓. 5n + 2 = 302 gives n = 60 columns, so there are 2 × 60 = 120 unit squares. (60 is the length, not the number of squares.) |
| **27** | Part 9 · Mixed · Ch5 + Ch6 · Hard | **(A)** 196 $cm^2$ | ***Hint:*** *The perpendicular from the centre bisects each chord. Find each chord's distance from the centre.* 16 cm chord: half is 8, distance = √(100 − 64) = 6 cm. 12 cm chord: half is 6, distance = √(100 − 36) = 8 cm. On opposite sides, the chords are 6 + 8 = 14 cm apart. Area = ½ × (16 + 12) × 14 = 196 $cm^2$. (28 $cm^2$ uses a height of 2, as if both chords were on the same side.) |
| **28** | Part 9 · Mixed · Ch4 + Ch5 · Hard | **(A)** 16 | ***Hint:*** *Its diagonals PQ and RS lie on the two axes.* P = (0, 1), Q = (0, −7), so PQ = 8. R = (−1/2, 0), S = (7/2, 0), so RS = 4. The diagonals are perpendicular (they lie on the axes), so area = ½ × 8 × 4 = 16. (32 forgets the ½.) |
| **29** | Part 9 · Mixed · Ch8 + Ch10 · Hard | **(B)** 4/25 | ***Hint:*** *Multiples of both 4 and 6 are multiples of 12.* Multiples of 4 up to 50: 12. Multiples of 12: 12, 24, 36, 48 — 4 of them. So 12 − 4 = 8 numbers: 8/50 = 4/25. |
| **30** | Part 9 · Mixed · Ch3 + Ch9 · Hard | **(A)** ₹3,900 | ***Hint:*** *The amounts owed are ₹60,000, ₹55,000, …, ₹5,000.* The loan lasts 12 months. Interest = 1% of (60,000 + 55,000 + … + 5,000) = 1% of 5,000 × (12 + 11 + … + 1) = 1% of 5,000 × 78 = ₹3,900. (₹7,200 charges 1% on the full ₹60,000 every month.) |

# ARYABHATTA MATHS OLYMPIAD

Level 2

## PRACTICE PAPER-9

# CLASS 9

| | |
|---|---|
| **Total Questions** | 30 multiple-choice questions |
| **Level** | Level-2 (final round): every question is Hard |
| **Time** | 45 minutes |
| **Arrangement** | Parts by chapter; the last Part has mixed problems joining two chapters |
| **Negative Marking** | No negative marking |
| **Calculator** | Not allowed |
| **Chapters** | Number System · Fractions, Decimals & Percentages · Algebra · Geometry · Mensuration · Data Handling, Statistics & Probability · Applied Mathematics · Mathematical Reasoning |

### Instructions

- Each question has four options, (A) to (D), with exactly one correct answer.
- The questions are grouped in Parts by chapter. The last Part has mixed problems, each joining two chapters.
- Every question is at Level-2 (Hard) standard: expect several steps or a key insight.
- There is no negative marking. Calculators are not allowed; use a separate sheet for rough work.
- Figures are not necessarily drawn to scale unless stated.

**For teachers**

- Figures are part of the questions and work when printed in black and white.
- Parts follow the AMO syllabus chapters. Chapters 2 and 7 are not part of the Class 9 syllabus, so the chapter numbers shown in the Part headings have gaps.
- The answer sheet and the answer key are at the end. Remove the answer key before the paper is attempted.

## Part 1 — Number System (Syllabus Chapter 1)

**Q1.** The decimal expansion of 1/13 repeats a block of digits. What is the sum of the digits in one repeating block?

**(A)** 30 **(B)** 24 **(C)** 36 **(D)** 27

**Q2.** In the square-root spiral, the first right triangle has legs 1 and 1, and the k-th triangle has hypotenuse √(k + 1). Which is the first triangle whose hypotenuse is longer than 5?

**(A)** 24th **(B)** 5th **(C)** 26th **(D)** 25th

## Part 2 — Fractions, Decimals & Percentages (Syllabus Chapter 3)

**Q3.** A clinic has 10,000 tablets. Plan P uses 1,500 tablets each month. Plan Q uses 15% of the tablets left each month. After 2 months, which plan leaves more tablets, and how many more?

**(A)** Plan P, by 225 **(B)** Plan Q, by 225

**(C)** Both leave the same **(D)** Plan Q, by 150

**Q4.** On a train, 40% of passengers pay ₹500, 35% pay ₹800 and the rest pay ₹1,200. What is the average fare per passenger?

**(A)** ₹833.33 **(B)** ₹780 **(C)** ₹800 **(D)** ₹750

## Part 3 — Algebra (Syllabus Chapter 4)

**Q5.** If $x + y + z = 0$ and $xy + yz + zx$ is not 0, what is $(x^2 + y^2 + z^2) \div (xy + yz + zx)$?

**(A)** −2 **(B)** 2 **(C)** −1 **(D)** 0

**Q6.** A rectangle made of algebra tiles has area $x^2 + 5x + 6$, and one side is $x + 2$. What is its perimeter?

**(A)** $2x + 5$ **(B)** $4x + 12$ **(C)** $4x + 10$ **(D)** $x^2 + 5x + 6$

**Q7.** A straight line passes through (1, −1) and is parallel to $y = 3x + 2$. Where does it cross the x-axis?

**(A)** (−4/3, 0) **(B)** (1/3, 0) **(C)** (2/3, 0) **(D)** (4/3, 0)

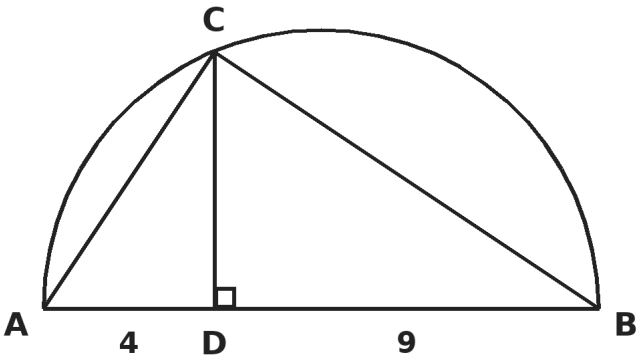
**Q8.** In a geometric progression of positive terms, the product of the 2nd and 6th terms is 144. What is the 4th term?

**(A)** 72 **(B)** 24 **(C)** 6 **(D)** 12

## Part 4 — Geometry (Syllabus Chapter 5)

**Q9.** Which of these points is NOT the same distance from A(2, −1) and B(−4, 3)?

**(A)** (0, 3) **(B)** (1, 4) **(C)** (−3, −2) **(D)** (3, 7)




**Q10.** AB is a diameter of a semicircle. D is on AB with AD = 4 cm and DB = 9 cm. The perpendicular to AB at D meets the semicircle at C. What is CD?

**(A)** 6.5 cm **(B)** 5 cm **(C)** 6 cm **(D)** 36 cm

**Q11.** A chord of a circle of radius 10 cm subtends a right angle at the centre. How far is the chord from the centre?

**(A)** 5√2 cm **(B)** 10 cm **(C)** 5 cm **(D)** 10√2 cm

**Q12.** In triangle ABC, AB = AC and angle A = 40°. The bisector of angle B meets AC at D. What is angle BDC?

**(A)** 70° **(B)** 105° **(C)** 75° **(D)** 35°

## Part 5 — Mensuration (Syllabus Chapter 6)

**Q13.** Circles of radius 7 cm are drawn with centres at the corners of an equilateral triangle of side 14 cm. What area of the triangle is not covered by the circles? (Take $\pi = 22/7$.)

**(A)** $49\sqrt{3} - 154$ cm² **(B)** $98\sqrt{3} - 77$ cm²

**(C)** 77 cm² **(D)** $49\sqrt{3} - 77$ cm²

**Q14.** A rectangular sheet 22 cm by 14 cm is rolled so that the 22 cm side forms the circular rim, making a cylinder of height 14 cm. What is its volume? (Take $\pi = 22/7$.)

**(A)** 539 cm³ **(B)** 343 cm³ **(C)** 1,078 cm³ **(D)** 154 cm³

**Q15.** A pyramid is cut from a cube of edge 6 cm. Its base is one face of the cube and its apex is the centre of the opposite face. What is the total surface area of the pyramid?

**(A)** $36 + 72\sqrt{5}$ cm² **(B)** $36 + 36\sqrt{5}$ cm² **(C)** 144 cm² **(D)** $36 + 18\sqrt{5}$ cm²

## Part 6 — Data Handling, Statistics & Probability (Syllabus Chapter 8)

**Q16.** A bag has 3 red balls and 2 blue balls. Kavya draws a ball, notes its colour, puts it back, and draws again. What is the probability that the two balls are of different colours?

**(A)** 6/25 **(B)** 13/25 **(C)** 12/25 **(D)** 1/2

**Q17.** The mean age of 8 people is 30 years. One person leaves and the mean of the remaining 7 becomes 28 years. A new person then joins, and the mean of the 8 becomes 29 years. How much younger is the new person than the one who left?

**(A)** 2 years **(B)** 16 years **(C)** 8 years **(D)** 6 years

**Q18.** Bag A has 2 red and 3 blue balls; bag B has 4 red and 1 blue. Pratik tosses a coin: heads he picks from A, tails from B. Then he draws one ball. What is the probability it is red?

**(A)** 1/2 **(B)** 3/5 **(C)** 6/25 **(D)** 2/5

## Part 7 — Applied Mathematics (Syllabus Chapter 9)

**Q19.** Gym A charges a ₹2,000 joining fee and ₹500 per month. Gym B charges ₹800 per month and no fee. From which month on is the total paid at Gym A less than at Gym B?

**(A)** 6th month **(B)** 7th month **(C)** 4th month **(D)** 9th month

**Q20.** A lab mixes a 30% acid solution with a 70% acid solution to make 50 L of 54% acid. How many litres of the 70% solution are used?

**(A)** 30 L **(B)** 20 L **(C)** 25 L **(D)** 27 L

**Q21.** A scholarship pays ₹1,000 in year 1, and each year it is 10% more than the year before. In which year does it first pay more than ₹1,500?

**(A)** Year 5 **(B)** Year 6 **(C)** Year 7 **(D)** Year 4

## Part 8 — Mathematical Reasoning (Syllabus Chapter 10)

**Q22.** Which statement has a converse that is also true?

**(A)** If a triangle is equilateral, each of its angles is 60°.

**(B)** If a whole number ends in 0, it is even.

**(C)** If a quadrilateral is a rectangle, its diagonals are equal.

**(D)** If $x = 2$, then $x^2 = 4$.

**Q23.** An algorithm sorts a list by repeatedly swapping two neighbours that are in the wrong order. What is the least number of swaps needed to sort 5, 4, 3, 2, 1 into 1, 2, 3, 4, 5?

**(A)** 5 **(B)** 4 **(C)** 10 **(D)** 20

**Q24.** An algorithm starts with the number 1 and repeats the step "multiply by 3, then subtract 1". It stops as soon as the number is more than 1,000. How many steps does it take?

**(A)** 7 **(B)** 6 **(C)** 8 **(D)** 5

**Q25.** Which pair of triangles disproves the claim "Two triangles with equal areas are congruent"?

**(A)** Two triangles with sides 3, 4 and 5 cm

**(B)** Two equilateral triangles of side 2 cm

**(C)** A right triangle and its mirror image

**(D)** Base 6 cm, height 4 cm and base 8 cm, height 3 cm

## Part 9 — Mixed Problems (each joins two chapters)

**Q26.** The triangular numbers are 1, 3, 6, 10, 15, … (the nth one is 1 + 2 + … + n). How many of the first 20 triangular numbers are even?

**(A)** 10 **(B)** 5 **(C)** 12 **(D)** 8

**Q27.** The triangle with vertices (0, 0), (6, 0) and (0, 8) is rotated once about the y-axis. What is the volume of the cone formed?

**(A)** 128π **(B)** 48π **(C)** 96π **(D)** 160π

**Q28.** The lines $y = 2x + 3$, $y = -3x + 8$ and $y = kx - 1$ all pass through one point. Where does the third line cross the x-axis?

**(A)** (1/4, 0) **(B)** (1/6, 0) **(C)** (−1/6, 0) **(D)** (6, 0)

**Q29.** Four letters are put at random into four addressed envelopes, one letter each. What is the probability that exactly three letters are in the right envelopes?

**(A)** 1/24 **(B)** 1/6 **(C)** 0 **(D)** 1/8

**Q30.** For ₹50,000 over 2 years, Option A pays 10% simple interest per year and Option B pays 9.5% compound interest per year. Which gives more interest, and by how much?

**(A)** Option B, by ₹48.75

**(B)** Both give the same interest

**(C)** Option A, by ₹500

**(D)** Option A, by ₹48.75

## Answer Sheet AMO Level 2 · Class 9 · Practice Paper 9

**Name:** **School:** **Roll No.:**

*Shade or tick one circle for each question.*

| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1. | A | B | C | D | 11. | A | B | C | D | 21. | A | B | C | D |
| 2. | A | B | C | D | 12. | A | B | C | D | 22. | A | B | C | D |
| 3. | A | B | C | D | 13. | A | B | C | D | 23. | A | B | C | D |
| 4. | A | B | C | D | 14. | A | B | C | D | 24. | A | B | C | D |
| 5. | A | B | C | D | 15. | A | B | C | D | 25. | A | B | C | D |
| 6. | A | B | C | D | 16. | A | B | C | D | 26. | A | B | C | D |
| 7. | A | B | C | D | 17. | A | B | C | D | 27. | A | B | C | D |
| 8. | A | B | C | D | 18. | A | B | C | D | 28. | A | B | C | D |
| 9. | A | B | C | D | 19. | A | B | C | D | 29. | A | B | C | D |
| 10. | A | B | C | D | 20. | A | B | C | D | 30. | A | B | C | D |

**For the parent or teacher**

**Correct: _____ / 30**

# Answer Key and Solutions AMO Level 2 · Class 9 · Practice Paper 9

## Quick answers

| | | | | | |
|---|---|---|---|---|---|
| 1 **D** | 6 **C** | 11 **A** | 16 **C** | 21 **B** | 26 **A** |
| 2 **D** | 7 **D** | 12 **C** | 17 **C** | 22 **A** | 27 **C** |
| 3 **B** | 8 **D** | 13 **D** | 18 **B** | 23 **C** | 28 **B** |
| 4 **B** | 9 **A** | 14 **A** | 19 **B** | 24 **A** | 29 **C** |
| 5 **A** | 10 **C** | 15 **B** | 20 **A** | 25 **D** | 30 **D** |

## Solutions

| Q | Part · level | Answer | Why? |
|---|---|---|---|
| **1** | Part 1 · Ch1 · Hard | **(D)** 27 | ***Hint:*** *Divide 1 by 13 until the remainder 1 comes back.* 1/13 = 0.076923 076923..., a block of 6 digits. 0 + 7 + 6 + 9 + 2 + 3 = 27. |
| **2** | Part 1 · Ch1 · Hard | **(D)** 25th | ***Hint:*** *$\sqrt{(k+1)} > 5$ means $k + 1 > 25$.* $\sqrt{(k+1)} > 5$ gives $k + 1 > 25$, so $k > 24$. The 24th triangle has hypotenuse $\sqrt{25} = 5$ exactly; the 25th has $\sqrt{26} > 5$. |
| **3** | Part 2 · Ch3 · Hard | **(B)** Plan Q, by 225 | ***Hint:*** *Plan Q multiplies by 0.85 each month.* P: 10,000 − 3,000 = 7,000. Q: 10,000 × 0.85 × 0.85 = 7,225. Plan Q leaves 225 more, because its second-month use is only 15% of 8,500. |
| **4** | Part 2 · Ch3 · Hard | **(B)** ₹780 | ***Hint:*** *The rest is 25%.* Average = 0.40 × 500 + 0.35 × 800 + 0.25 × 1,200 = 200 + 280 + 300 = ₹780. (₹833.33 averages the three fares without weights.) |
| **5** | Part 3 · Ch4 · Hard | **(A)** −2 | ***Hint:*** *Square $x + y + z$.* $(x + y + z)^2 = x^2 + y^2 + z^2 + 2(xy + yz + zx) = 0$, so $x^2 + y^2 + z^2 = -2(xy + yz + zx)$. The ratio is −2. |
| **6** | Part 3 · Ch4 · Hard | **(C)** 4x + 10 | ***Hint:*** *Factorise the area.* $x^2 + 5x + 6 = (x + 2)(x + 3)$, so the other side is $x + 3$. Perimeter = $2(x + 2 + x + 3) = 4x + 10$. |
| **7** | Part 3 · Ch4 · Hard | **(D)** (4/3, 0) | ***Hint:*** *Parallel lines have the same slope.* The line is $y = 3x + b$ with $-1 = 3 + b$, so $b = -4$ and $y = 3x - 4$. At $y = 0$, $x = 4/3$. ((1/3, 0) comes from taking $b = -1$, the y-value of the point; (−4/3, 0) has the sign wrong.) |
| **8** | Part 3 · Ch4 · Hard | **(D)** 12 | ***Hint:*** *Write the terms as $ar$ and $ar^5$.* $ar \times ar^5 = a^2r^6 = (ar^3)^2 = 144$. The 4th term is $ar^3 = 12$ (positive). |
| **9** | Part 4 · Ch5 · Hard | **(A)** (0, 3) | ***Hint:*** *Compare squared distances to A and to B.* (1, 4): 1 + 25 = 26 and 25 + 1 = 26 ✓. (−3, −2): 25 + 1 and 1 + 25 ✓. (3, 7): 1 + 64 and 49 + 16 ✓. (0, 3): 4 + 16 = 20 but 16 + 0 = 16 ×. The first three lie on the perpendicular bisector of AB. |
| **10** | Part 4 · Ch5 · Hard | **(C)** 6 cm | ***Hint:*** *Join the centre O to C. OC is a radius and OD = OA − AD.* AB = 13, so the radius is 6.5 cm. OD = 6.5 − 4 = 2.5 cm. In right triangle ODC: CD = $\sqrt{(6.5^2 - 2.5^2)} = \sqrt{36} = 6$ cm. |

| Q | Part · level | Answer | Why? |
|---|---|---|---|
| **11** | Part 4 · Ch5 · Hard | **(A)** $5\sqrt{2}$ cm | ***Hint:*** *The chord is the hypotenuse of a right isosceles triangle with legs 10.* The chord = $10\sqrt{2}$ cm. The perpendicular from the centre bisects it, so the distance is $\sqrt{(10^2 - (5\sqrt{2})^2)} = \sqrt{50} = 5\sqrt{2}$ cm. |
| **12** | Part 4 · Ch5 · Hard | **(C)** 75° | ***Hint:*** *Find angle B first.* Angles B and C are each $(180° - 40°)/2 = 70°$. Angle DBC = 35°. In triangle BDC: angle BDC = 180° − 35° − 70° = 75°. |
| **13** | Part 5 · Ch6 · Hard | **(D)** $49\sqrt{3} - 77$ cm² | ***Hint:*** *Each corner holds a 60° sector.* Triangle area = $(\sqrt{3}/4) \times 14^2 = 49\sqrt{3}$. Three 60° sectors make half a circle: $\frac{1}{2} \times 22/7 \times 49 = 77$. Uncovered = $49\sqrt{3} - 77$ cm² (about 7.9 cm²). |
| **14** | Part 5 · Ch6 · Hard | **(A)** 539 cm³ | ***Hint:*** *The circumference is 22 cm.* $2\pi r = 22$ gives r = 3.5 cm. Volume = $22/7 \times 3.5^2 \times 14 = 539$ cm³. (343 cm³ comes from rolling the other way.) |
| **15** | Part 5 · Ch6 · Hard | **(B)** $36 + 36\sqrt{5}$ cm² | ***Hint:*** *Each sloping face is a triangle with base 6 cm. Find its slant height from the height 6 cm and half the base side, 3 cm.* Slant height = $\sqrt{(6^2 + 3^2)} = \sqrt{45} = 3\sqrt{5}$ cm. Each triangular face = $\frac{1}{2} \times 6 \times 3\sqrt{5} = 9\sqrt{5}$ cm², and there are 4: $36\sqrt{5}$ cm². Adding the square base 36 cm² gives $36 + 36\sqrt{5}$ cm². ($36 + 18\sqrt{5}$ uses only two triangles.) |
| **16** | Part 6 · Ch8 · Hard | **(C)** 12/25 | ***Hint:*** *Draw a tree: red then blue, or blue then red.* P(red then blue) = $3/5 \times 2/5 = 6/25$ and P(blue then red) = $2/5 \times 3/5 = 6/25$. Total = 12/25. (6/25 counts only one order; 13/25 is the probability of the same colour.) |
| **17** | Part 6 · Ch8 · Hard | **(C)** 8 years | ***Hint:*** *Work with totals: 8 × 30, then 7 × 28, then 8 × 29.* Totals: 240, then 196, so the person who left is 44. Then 8 × 29 = 232, so the new person is 232 − 196 = 36. The new person is 44 − 36 = 8 years younger. (2 years compares the two averages.) |
| **18** | Part 6 · Ch8 · Hard | **(B)** 3/5 | ***Hint:*** *Use a tree with two stages: bag, then ball.* P(red) = $\frac{1}{2} \times 2/5 + \frac{1}{2} \times 4/5 = 1/5 + 2/5 = 3/5$. (6/25 multiplies 2/5 by 4/5 instead of adding the branches.) |
| **19** | Part 7 · Ch9 · Hard | **(B)** 7th month | ***Hint:*** *Compare 2,000 + 500m with 800m.* 2,000 + 500m < 800m gives 300m > 2,000, so m > 6.67. After 6 months A totals ₹5,000 and B ₹4,800; after 7 months A totals ₹5,500 and B ₹5,600. So from the 7th month. |
| **20** | Part 7 · Ch9 · Hard | **(A)** 30 L | ***Hint:*** *Let the amounts be x and y with x + y = 50.* $0.3x + 0.7y = 0.54 \times 50 = 27$ and $x + y = 50$. So $15 + 0.4y = 27$, y = 30 L of the 70% solution and x = 20 L of the 30% solution. |
| **21** | Part 7 · Ch9 · Hard | **(B)** Year 6 | ***Hint:*** *Year n pays* $1{,}000 \times 1.1^{n-1}$. Payments: 1,000, 1,100, 1,210, 1,331, 1,464.10, 1,610.51. Year 6 is the first above ₹1,500. (Adding a flat ₹100 a year would give exactly ₹1,500 in year 6, not more.) |
| **22** | Part 8 · Ch10 · Hard | **(A)** If a triangle is equilateral, each of its angles is 60°. | ***Hint:*** *Swap the parts of each statement and test it.* "If each angle is 60°, the triangle is equilateral" is true. The others fail: 4 is even but does not end in 0; an isosceles trapezium has equal diagonals; $x = -2$ gives $x^2 = 4$. |
| **23** | Part 8 · Ch10 · Hard | **(C)** 10 | ***Hint:*** *Each swap fixes exactly one pair that is out of order.* Every pair of the 5 numbers is out of order: 4 + 3 + 2 + 1 = 10 pairs. Each adjacent swap fixes exactly one pair, so 10 swaps are needed (and enough). |

| Q | Part · level | Answer | Why? |
|---|---|---|---|
| **24** | Part 8 · Ch10 · Hard | **(A)** 7 | ***Hint:*** *Write down the numbers one step at a time.* 1 → 2 → 5 → 14 → 41 → 122 → 365 → 1,094. After 6 steps the number is 365; the 7th step gives 1,094, which is more than 1,000. So 7 steps. |
| **25** | Part 8 · Ch10 · Hard | **(D)** Base 6 cm, height 4 cm and base 8 cm, height 3 cm | ***Hint:*** *You need equal areas but different shapes.* The triangles with base 6 cm, height 4 cm and base 8 cm, height 3 cm both have area 12 $cm^2$, but their bases differ, so they cannot be congruent. The other pairs are congruent, so they agree with the claim. |
| **26** | Part 9 · Mixed · Ch4 + Ch10 · Hard | **(A)** 10 | ***Hint:*** *Look at the pattern of odd and even in the first eight: 1, 3, 6, 10, 15, 21, 28, 36.* The pattern of parity repeats every 4 terms: odd, odd, even, even. In each block of 4 there are 2 even numbers, so the first 20 contain 5 × 2 = 10 even triangular numbers (the 3rd, 4th, 7th, 8th, …, 19th, 20th). |
| **27** | Part 9 · Mixed · Ch5 + Ch6 · Hard | **(C)** 96π | ***Hint:*** *The leg on the y-axis is the height; the other leg is the radius.* Radius 6 and height 8: V = ⅓ × π × 36 × 8 = 96π. (128π comes from rotating about the x-axis.) |
| **28** | Part 9 · Mixed · Ch4 + Ch5 · Hard | **(B)** (1/6, 0) | ***Hint:*** *Find where the first two lines meet, then find k.* 2x + 3 = −3x + 8 gives x = 1, y = 5. The third line passes through (1, 5): 5 = k − 1, so k = 6. Then 6x − 1 = 0 gives x = 1/6, so it crosses at (1/6, 0). ((6, 0) uses k itself.) |
| **29** | Part 9 · Mixed · Ch8 + Ch10 · Hard | **(C)** 0 | ***Hint:*** *If three letters are right, where must the fourth go?* If three letters are in their own envelopes, the only envelope left is the fourth letter's own, so it is right too. Exactly three correct is impossible: probability 0. |
| **30** | Part 9 · Mixed · Ch3 + Ch9 · Hard | **(D)** Option A, by ₹48.75 | ***Hint:*** *Option B: 50,000 × $1.095^2$ − 50,000.* A: 2 × 5,000 = ₹10,000. B: 50,000 × 1.199025 − 50,000 = ₹9,951.25. Option A gives ₹48.75 more: compounding does not make up for the 0.5% lower rate in only 2 years. |

# ARYABHATTA MATHS OLYMPIAD

Level 2

## PRACTICE PAPER-10

# CLASS 9

| | |
|---|---|
| **Total Questions** | 30 multiple-choice questions |
| **Level** | Level-2 (final round): every question is Hard |
| **Time** | 45 minutes |
| **Arrangement** | Parts by chapter; the last Part has mixed problems joining two chapters |
| **Negative Marking** | No negative marking |
| **Calculator** | Not allowed |
| **Chapters** | Number System · Fractions, Decimals & Percentages · Algebra · Geometry · Mensuration · Data Handling, Statistics & Probability · Applied Mathematics · Mathematical Reasoning |

### Instructions

- Each question has four options, (A) to (D), with exactly one correct answer.
- The questions are grouped in Parts by chapter. The last Part has mixed problems, each joining two chapters.
- Every question is at Level-2 (Hard) standard: expect several steps or a key insight.
- There is no negative marking. Calculators are not allowed; use a separate sheet for rough work.
- Figures are not necessarily drawn to scale unless stated.

**For teachers**

- Figures are part of the questions and work when printed in black and white.
- Parts follow the AMO syllabus chapters. Chapters 2 and 7 are not part of the Class 9 syllabus, so the chapter numbers shown in the Part headings have gaps.
- The answer sheet and the answer key are at the end. Remove the answer key before the paper is attempted.

## Part 1 — Number System (Syllabus Chapter 1)

**Q1.** Write the recurring decimal 1.272727… (27 repeating) as a fraction in lowest terms.

**(A)** 127/99 **(B)** 63/50 **(C)** 14/11 **(D)** 13/11

**Q2.** p is irrational and q is a non-zero rational number. Which of these must be irrational?

**(A)** p × p **(B)** p − p **(C)** p ÷ p **(D)** p + q

## Part 2 — Fractions, Decimals & Percentages (Syllabus Chapter 3)

**Q3.** A town of 60,000 people grows by 5% in one year and then shrinks by 5% in the next. What is its population after the two years?

**(A)** 59,850 **(B)** 60,000 **(C)** 59,700 **(D)** 60,150

**Q4.** A grocer mixes rice costing ₹40 per kg with rice costing ₹64 per kg so that the mixture costs ₹50 per kg. In what ratio (cheaper : dearer) are they mixed?

**(A)** 5 : 7 **(B)** 7 : 5 **(C)** 3 : 2 **(D)** 2 : 3

## Part 3 — Algebra (Syllabus Chapter 4)

**Q5.** If $a + b + c = 9$ and $a^2 + b^2 + c^2 = 35$, what is $ab + bc + ca$?

**(A)** 46 **(B)** 23 **(C)** 22 **(D)** 58

**Q6.** Two numbers x and y satisfy $x^2 - y^2 = 45$ and $x + y = 9$. What is $x^3 - y^3$?

**(A)** 335 **(B)** 351 **(C)** 45 **(D)** 125

**Q7.** The equations $2x + 3y = 7$ and $ax + 9y = b$ have infinitely many solutions. What is $a + b$?

**(A)** 15 **(B)** 21 **(C)** 27 **(D)** 6

**Q8.** For how many whole numbers n is 1 + 2 + … + n greater than 400 and less than 450?

**(A)** 1 **(B)** 3 **(C)** 2 **(D)** 0

## Part 4 — Geometry (Syllabus Chapter 5)

**Q9.** A = (−3, 5) and B = (5, −1). M = (2, 4) is the midpoint of AC. What is the length of BC?

**(A)** 10 **(B)** $4\sqrt{5}$ **(C)** 6 **(D)** $2\sqrt{5}$

**Q10.** ABCD is a cyclic quadrilateral with angle A = (2x + 10)°, angle B = 3x° and angle C = (2x + 30)°. What is angle D?

**(A)** 75° **(B)** 105° **(C)** 80° **(D)** 100°

**Q11.** Where is the circumcentre of the triangle with vertices (0, 0), (8, 0) and (2, 2)?

**(A)** (4, 2) **(B)** (4, −2) **(C)** (10/3, 2/3) **(D)** (4, 0)

**Q12.** The medians of triangle ABC meet at the centroid G. The area of triangle ABC is 54 $cm^2$. What is the area of triangle AGB?

**(A)** 18 $cm^2$ **(B)** 27 $cm^2$ **(C)** 9 $cm^2$ **(D)** 36 $cm^2$

## Part 5 — Mensuration (Syllabus Chapter 6)

**Q13.** A rhombus has side 13 cm and one diagonal 24 cm. What is its area?

**(A)** 156 $cm^2$ **(B)** 240 $cm^2$ **(C)** 120 $cm^2$ **(D)** 60 $cm^2$

**Q14.** A cone and a hemisphere have the same base radius, 7 cm, and the same volume. What is the height of the cone?

**(A)** 14 cm **(B)** 7 cm **(C)** 21 cm **(D)** 28 cm

**Q15.** The circumference of a circle is 30 cm more than its diameter. What is its radius? (Take $\pi = 22/7$.)

**(A)** 14 cm **(B)** 3.5 cm **(C)** 10.5 cm **(D)** 7 cm

## Part 6 — Data Handling, Statistics & Probability (Syllabus Chapter 8)

**Q16.** A family has three children, and boys and girls are equally likely. We are told the family has at least one girl. What is the probability that all three children are girls?

**(A)** 1/8 **(B)** 1/7 **(C)** 1/3 **(D)** 1/4

| Value | 2 | 4 | 6 | 8 |
|---|---|---|---|---|
| Frequency | 3 | p | 5 | 2 |

**Q17.** The mean of the data in the table is 5. What is p?

**(A)** 2 **(B)** 3 **(C)** 4 **(D)** 5

| School | Students | Cricket | Football | Other |
|---|---|---|---|---|
| A | 300 | 50% | 30% | 20% |
| B | 500 | 30% | 45% | 25% |

**Q18.** The table shows how the students of two schools choose a sport, as percentages of each school. What percentage of all 800 students choose cricket?

**(A)** 40% **(B)** 35% **(C)** 45% **(D)** 37.5%

## Part 7 — Applied Mathematics (Syllabus Chapter 9)

**Q19.** A machine needs 15 minutes to set up and then fills 120 bottles per hour. Work starts at 9:00 AM. When will 450 bottles be filled?

**(A)** 1:00 PM **(B)** 12:45 PM **(C)** 12:30 PM **(D)** 1:15 PM

**Q20.** Kartik travels 20 km in 3 hours, walking part of the way at 4 km/h and cycling the rest at 12 km/h. How far does he cycle?

**(A)** 12 km **(B)** 8 km **(C)** 10 km **(D)** 15 km

**Q21.** A loan is repaid in 20 monthly instalments. The first is ₹30,000 and each one is ₹1,000 less than the one before. What is the total repaid?

**(A)** ₹6,00,000 **(B)** ₹4,20,000 **(C)** ₹3,90,000 **(D)** ₹4,10,000

## Part 8 — Mathematical Reasoning (Syllabus Chapter 10)

**Q22.** Statement: "If a quadrilateral is a rhombus, then its diagonals are perpendicular." Which option gives its converse correctly, with whether the converse is true?

**(A)** If the diagonals are perpendicular, it is a rhombus: true

**(B)** If the diagonals are perpendicular, it is a rhombus: false

**(C)** If it is not a rhombus, its diagonals are not perpendicular: false

**(D)** If the diagonals are not perpendicular, it is not a rhombus: true

**Q23.** Algorithm: add a number to its reverse, and repeat until the result reads the same forwards and backwards. Starting with 87, how many additions are needed?

**(A)** 3 **(B)** 5 **(C)** 2 **(D)** 4

**Q24.** A square of side 9 cm is cut into 9 equal squares and the middle one is removed. Then each of the remaining 8 squares is treated the same way. What total area has been removed?

**(A)** 9 $cm^2$ **(B)** 16 $cm^2$ **(C)** 17 $cm^2$ **(D)** 64 $cm^2$

**Q25.** Which theorem is a generalisation of "The angle in a semicircle is a right angle"?

**(A)** Opposite angles of a cyclic quadrilateral are equal.

**(B)** Equal chords are equidistant from the centre.

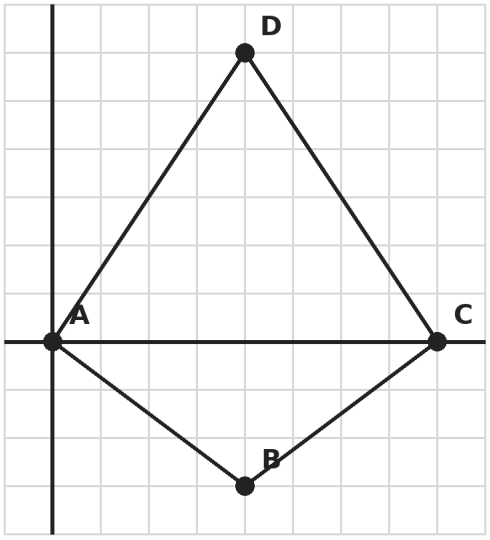
**(C)** The angle an arc subtends at the centre is twice the angle it subtends at any point on the rest of the circle.

**(D)** A diameter is the longest chord of a circle.

## Part 9 — Mixed Problems (each joins two chapters)

**Q26.** On an n × n grid of unit squares, Ishaan counts all squares of every size and gets 204. What is n?

**(A)** 7 **(B)** 9 **(C)** 6 **(D)** 8




**Q27.** What is the area of the quadrilateral with vertices A(0, 0), B(4, −3), C(8, 0) and D(4, 6)?

**(A)** 72 **(B)** 36 **(C)** 24 **(D)** 48

**Q28.** The line y = (3/4)x + 3 crosses the x-axis at P and the y-axis at Q. What is the shortest distance from the origin O to the line PQ?

**(A)** 5 **(B)** 7/2 **(C)** 12/5 **(D)** 3

**Q29.** A three-digit number is formed using only the digits 1, 2 and 3, and digits may repeat. All such numbers are equally likely. What is the probability that its digits are all different?

**(A)** 1/3 **(B)** 2/9 **(C)** 1/9 **(D)** 2/3

**Q30.** Ankit invests ₹20,000 at 10% compound interest per year. After one year he adds ₹5,000 more to the account. How much is in the account at the end of the second year?

**(A)** ₹29,200 **(B)** ₹30,250 **(C)** ₹29,000 **(D)** ₹29,700

## Answer Sheet AMO Level 2 · Class 9 · Practice Paper 10

**Name:** **School:** **Roll No.:**

*Shade or tick one circle for each question.*

| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1. | A | B | C | D | 11. | A | B | C | D | 21. | A | B | C | D |
| 2. | A | B | C | D | 12. | A | B | C | D | 22. | A | B | C | D |
| 3. | A | B | C | D | 13. | A | B | C | D | 23. | A | B | C | D |
| 4. | A | B | C | D | 14. | A | B | C | D | 24. | A | B | C | D |
| 5. | A | B | C | D | 15. | A | B | C | D | 25. | A | B | C | D |
| 6. | A | B | C | D | 16. | A | B | C | D | 26. | A | B | C | D |
| 7. | A | B | C | D | 17. | A | B | C | D | 27. | A | B | C | D |
| 8. | A | B | C | D | 18. | A | B | C | D | 28. | A | B | C | D |
| 9. | A | B | C | D | 19. | A | B | C | D | 29. | A | B | C | D |
| 10. | A | B | C | D | 20. | A | B | C | D | 30. | A | B | C | D |

**For the parent or teacher**

**Correct: _____ / 30**

# Answer Key and Solutions AMO Level 2 · Class 9 · Practice Paper 10

## Quick answers

| | | | | | |
|---|---|---|---|---|---|
| 1 **C** | 6 **A** | 11 **B** | 16 **B** | 21 **D** | 26 **D** |
| 2 **D** | 7 **C** | 12 **A** | 17 **A** | 22 **B** | 27 **B** |
| 3 **A** | 8 **C** | 13 **C** | 18 **D** | 23 **D** | 28 **C** |
| 4 **B** | 9 **D** | 14 **A** | 19 **A** | 24 **C** | 29 **B** |
| 5 **B** | 10 **A** | 15 **D** | 20 **A** | 25 **C** | 30 **D** |

## Solutions

| Q | Part · level | Answer | Why? |
|---|---|---|---|
| **1** | Part 1 · Ch1 · Hard | **(C)** 14/11 | ***Hint:*** *0.2727… = 27/99.* 1.2727… = 1 + 27/99 = 1 + 3/11 = 14/11. (127/99 treats the whole number part as part of the repeating block.) |
| **2** | Part 1 · Ch1 · Hard | **(D)** p + q | ***Hint:*** *Try p = √2 in each.* If p + q were rational, p = (p + q) − q would be rational, which is false; so p + q is irrational. But √2 × √2 = 2, p − p = 0 and p ÷ p = 1 are rational. |
| **3** | Part 2 · Ch3 · Hard | **(A)** 59,850 | ***Hint:*** *The factors are 1.05 and 0.95.* 60,000 × 1.05 × 0.95 = 60,000 × 0.9975 = 59,850. The 5% fall is taken on the larger number, so it does not cancel the rise. |
| **4** | Part 2 · Ch3 · Hard | **(B)** 7 : 5 | ***Hint:*** *Each kilogram of cheap rice saves ₹10 against ₹50; each kilogram of dear rice costs ₹14 extra.* Let the amounts be p kg and q kg. 40p + 64q = 50(p + q) gives 14q = 10p, so p : q = 14 : 10 = 7 : 5. |
| **5** | Part 3 · Ch4 · Hard | **(B)** 23 | ***Hint:*** *Expand $(a + b + c)^2$.* 81 = 35 + 2(ab + bc + ca), so ab + bc + ca = 46 ÷ 2 = 23. |
| **6** | Part 3 · Ch4 · Hard | **(A)** 335 | ***Hint:*** *$x^2 - y^2 = (x + y)(x - y)$. Find x and y first.* (x + y)(x − y) = 45 gives x − y = 5, so x = 7 and y = 2. $x^3 - y^3 = 343 - 8 = 335$. (Or $(x - y)(x^2 + xy + y^2) = 5 \times 67 = 335$.) (125 is $(x - y)^3$.) |
| **7** | Part 3 · Ch4 · Hard | **(C)** 27 | ***Hint:*** *The second equation must be a multiple of the first.* 9 = 3 × 3, so the second equation is 3 times the first: a = 6 and b = 21. a + b = 27. |
| **8** | Part 3 · Ch4 · Hard | **(C)** 2 | ***Hint:*** *The sum is n(n + 1)/2. Try n near 28.* n = 27 gives 378, n = 28 gives 406, n = 29 gives 435 and n = 30 gives 465. Only n = 28 and n = 29 work: 2 values. |
| **9** | Part 4 · Ch5 · Hard | **(D)** 2√5 | ***Hint:*** *Find C from the midpoint first.* C = (2 × 2 − (−3), 2 × 4 − 5) = (7, 3). BC = $\sqrt{(7 - 5)^2 + (3 + 1)^2} = \sqrt{20} = 2\sqrt{5}$. (10 is the length AB.) |
| **10** | Part 4 · Ch5 · Hard | **(A)** 75° | ***Hint:*** *Angles A and C are opposite.* A + C = 180°: 4x + 40 = 180, so x = 35. Angle B = 105°, and angle D = 180° − 105° = 75°. |

| Q | Part · level | Answer | Why? |
|---|---|---|---|
| **11** | Part 4 · Ch5 · Hard | **(B)** (4, −2) | ***Hint:*** *It lies on the perpendicular bisector of the side from (0, 0) to (8, 0), the line x = 4.* On x = 4, the point (4, y) must be the same distance from (0, 0) and (2, 2): $16 + y^2 = 4 + (y - 2)^2$, so 16 = 8 − 4y and y = −2. The circumcentre (4, −2) lies outside, as the triangle is obtuse. (10/3, 2/3) is the centroid. |
| **12** | Part 4 · Ch5 · Hard | **(A)** 18 cm² | ***Hint:*** *G is ⅓ of the way from each side to the opposite vertex.* The median from C splits ABC into two halves of 27 cm². G divides that median 2 : 1, so G is at ⅓ of the height of C above AB. Triangle AGB = ⅓ × 54 = 18 cm². |
| **13** | Part 5 · Ch6 · Hard | **(C)** 120 cm² | ***Hint:*** *The diagonals bisect each other at right angles.* Half the other diagonal = $\sqrt{(13^2 - 12^2)} = 5$, so it is 10 cm. Area = ½ × 24 × 10 = 120 cm². (240 forgets the ½.) |
| **14** | Part 5 · Ch6 · Hard | **(A)** 14 cm | ***Hint:*** *Set $\frac{1}{3}\pi r^2 h$ equal to $\frac{2}{3}\pi r^3$.* $\frac{1}{3}\pi r^2 h = \frac{2}{3}\pi r^3$ gives h = 2r = 14 cm. |
| **15** | Part 5 · Ch6 · Hard | **(D)** 7 cm | ***Hint:*** *Write $2\pi r - 2r = 30$.* $2r(\pi - 1) = 30$. With π = 22/7: 2r × 15/7 = 30, so r = 7 cm. |
| **16** | Part 6 · Ch8 · Hard | **(B)** 1/7 | ***Hint:*** *List the 8 outcomes and cross out the one with no girl.* Of the 8 equally likely outcomes, only BBB has no girl, leaving 7. Just one of these, GGG, has three girls. Probability = 1/7. (1/8 ignores the information that there is at least one girl.) |
| **17** | Part 6 · Ch8 · Hard | **(A)** 2 | ***Hint:*** *Mean = (sum of value × frequency) ÷ (sum of frequencies).* (6 + 4p + 30 + 16)/(10 + p) = 5 gives 52 + 4p = 50 + 5p, so p = 2. |
| **18** | Part 6 · Ch8 · Hard | **(D)** 37.5% | ***Hint:*** *Change each percentage into a number of students.* A: 50% of 300 = 150. B: 30% of 500 = 150. Total 300 of 800 = 37.5%. (40% averages 50% and 30% without weighting.) |
| **19** | Part 7 · Ch9 · Hard | **(A)** 1:00 PM | ***Hint:*** *450 ÷ 120 hours of filling, plus the set-up.* Filling time = 450 ÷ 120 = 3.75 h = 3 h 45 min. With 15 min set-up: 4 hours. 9:00 AM + 4 h = 1:00 PM. |
| **20** | Part 7 · Ch9 · Hard | **(A)** 12 km | ***Hint:*** *Let w and c be the distances walked and cycled.* w + c = 20 and w/4 + c/12 = 3, i.e. 3w + c = 36. Subtracting: 2w = 16, so w = 8 km and c = 12 km. |
| **21** | Part 7 · Ch9 · Hard | **(D)** ₹4,10,000 | ***Hint:*** *Total = 20 × 30,000 − 1,000 × (0 + 1 + … + 19).* 0 + 1 + … + 19 = 190. Total = 6,00,000 − 1,90,000 = ₹4,10,000. |
| **22** | Part 8 · Ch10 · Hard | **(B)** If the diagonals are perpendicular, it is a rhombus: false | ***Hint:*** *The converse swaps the "if" part and the "then" part. Think of a kite.* The converse is "If the diagonals are perpendicular, it is a rhombus". A kite with unequal sides has perpendicular diagonals but is not a rhombus, so the converse is false. The last option is the contrapositive (true, but not the converse). |
| **23** | Part 8 · Ch10 · Hard | **(D)** 4 | ***Hint:*** *87 + 78 = 165.* 87 + 78 = 165; 165 + 561 = 726; 726 + 627 = 1,353; 1,353 + 3,531 = 4,884, a palindrome. That is 4 additions. |
| **24** | Part 8 · Ch10 · Hard | **(C)** 17 cm² | ***Hint:*** *Step 1 removes a 3 cm square; step 2 removes eight 1 cm squares.* Step 1 removes 3 × 3 = 9 cm². Step 2 removes 8 squares of 1 cm², i.e. 8 cm². Total = 17 cm² (and 64 cm² remains). |

| Q | Part · level | Answer | Why? |
|---|---|---|---|
| **25** | Part 8 · Ch10 · Hard | **(C)** The angle an arc subtends at the centre is twice the angle it subtends at any point on the rest of the circle. | ***Hint:*** *A generalisation includes the original result as a special case.* For a semicircle, the arc subtends 180° at the centre, so the angle on the circle is half of it: 90°. So the "twice the angle" theorem contains the semicircle result. The cyclic-quadrilateral statement is false (opposite angles add up to 180°, they are not equal); the chord and diameter statements are true but different results. |
| **26** | Part 9 · Mixed · Ch4 + Ch10 · Hard | **(D)** 8 | ***Hint:*** *An $n \times n$ grid has $n^2$ unit squares, $(n-1)^2$ squares of side 2, and so on.* Total = $n^2 + (n-1)^2 + \ldots + 1^2$. For n = 7: 49 + 36 + 25 + 16 + 9 + 4 + 1 = 140. For n = 8: 140 + 64 = 204. So n = 8 (a chessboard). (n = 9 gives 285.) |
| **27** | Part 9 · Mixed · Ch5 + Ch6 · Hard | **(B)** 36 | ***Hint:*** *The diagonals AC and BD are perpendicular.* AC lies on the x-axis with length 8. BD lies on the line x = 4 with length 9. The diagonals are perpendicular, so area = ½ × 8 × 9 = 36. |
| **28** | Part 9 · Mixed · Ch4 + Ch5 · Hard | **(C)** 12/5 | ***Hint:*** *Find P and Q, then work out the area of triangle OPQ in two ways.* Q = (0, 3) and P = (−4, 0), so OQ = 3, OP = 4 and PQ = 5. Area of OPQ = ½ × 4 × 3 = 6 = ½ × 5 × d, so d = 12/5. (5 is PQ itself; 3 is OQ, which is not perpendicular to PQ.) |
| **29** | Part 9 · Mixed · Ch8 + Ch10 · Hard | **(B)** 2/9 | ***Hint:*** *There are 3 × 3 × 3 numbers in all.* Total = 27. With all digits different: 3 × 2 × 1 = 6 numbers (123, 132, …). Probability = 6/27 = 2/9. |
| **30** | Part 9 · Mixed · Ch3 + Ch9 · Hard | **(D)** ₹29,700 | ***Hint:*** *The ₹5,000 earns interest for only one year.* After year 1: 22,000. Add 5,000: 27,000. After year 2: 27,000 × 1.1 = ₹29,700. (₹30,250 lets the ₹5,000 grow for two years.) |